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AI-Based Diabetes Prediction and Classification

A thesis submitted by

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**A dissertation submitted in partial fulfillment of the requirements for the degree of Digilians 9
Months Diploma in Applied AI and Data Analytics**

Under supervision of

General Prof. Dr. Kamel A. El-Hadad

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Approval Sheet

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Abstract

Diabetes is a major global health concern, and early identification of individuals at risk can support timely intervention and improve health outcomes. This project presents GlucoGuide, an AI-powered web-based system designed for preliminary diabetes risk assessment and clinical decision support.

GlucoGuide combines a machine learning prediction model with an interactive website that collects demographic, clinical, lifestyle, and medical history data, including BMI, blood glucose, HbA1c, cholesterol, blood pressure, and insulin levels. The system also integrates Optical Character Recognition (OCR) to extract relevant laboratory values from uploaded medical reports, reducing manual data entry and improving the efficiency of the assessment process.

The collected data are processed and passed through a prediction API to the trained machine learning model, which generates an estimated diabetes risk assessment. The results are then presented through a clear and user-friendly web interface to support risk awareness and interpretation.

Overall, GlucoGuide demonstrates how Artificial Intelligence, Machine Learning, OCR, and web technologies can be integrated into a unified healthcare system to provide accessible and efficient preliminary diabetes risk assessment. The system is intended to support, not replace, professional medical diagnosis and clinical judgment.

Keywords: Diabetes Risk Assessment, Artificial Intelligence, Machine Learning, OCR, Clinical Decision Support, Healthcare, Web-Based System

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Nomenclature

Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
ML	Machine Learning
EDA	Exploratory Data Analysis
XAI	Explainable Artificial Intelligence
SVM	Support Vector Machine
RF	Random Forest
XGBoost	Extreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
MLP	Multi-Layer Perceptron
SMOTE	Synthetic Minority Over-sampling Technique
PCA	Principal Component Analysis
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-Agnostic Explanations
RAG	Retrieval-Augmented Generation
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve

BMI	Body Mass Index
HbA1c	Hemoglobin A1c
MLflow	Machine Learning Flow
API	Application Programming Interface
OCR	Optical Character Recognition
ROS	Random Oversampling
SMOTEENN	Synthetic Minority Over-sampling Technique with Edited Nearest Neighbours
CV	Cross-Validation

CHAPTER 1

INTRODUCTION

1.1 Overview

Artificial Intelligence (AI) and Machine Learning (ML) technologies have transformed many sectors over the past decade, particularly the healthcare industry. Modern healthcare systems increasingly utilize intelligent technologies to support disease prediction, risk assessment, monitoring, and clinical decision-making. Among the major chronic diseases worldwide is **Diabetes Mellitus**, a metabolic disorder characterized by elevated blood glucose levels resulting from insufficient insulin production, ineffective insulin utilization, or a combination of both.

Diabetes represents a major global health challenge because of its increasing prevalence and its potential to cause serious long-term complications, including cardiovascular disease, kidney disease, nerve damage, and vision impairment. Early identification of individuals at increased risk can support preventive actions and encourage timely clinical assessment.

Traditional diabetes assessment relies primarily on laboratory measurements, medical history, and clinical evaluation. Although these approaches remain essential, Machine Learning provides an additional opportunity to analyze multiple clinical and lifestyle-related variables simultaneously and identify complex patterns associated with different diabetes conditions.

Machine Learning models can utilize features such as age, gender, body measurements, HbA1c, fasting glucose, postprandial glucose, cholesterol, blood pressure, insulin level, physical activity, and family history of diabetes to generate automated predictions. However, high predictive performance alone is not sufficient for healthcare applications. A prediction system should also provide understandable explanations and present information in a form that can be used by patients and healthcare professionals. One of the major challenges in AI-based healthcare systems is therefore the **interpretability of machine learning predictions**. Many high-performing models operate as black boxes, making it difficult for users to understand why a particular prediction was generated. Explainable Artificial Intelligence (XAI),

particularly **SHAP (SHapley Additive exPlanations)**, can address this limitation by identifying the contribution of individual features to model predictions.

In addition to prediction and explainability, users may require reliable guidance after receiving a risk assessment. For this reason, the proposed system incorporates a **Retrieval-Augmented Generation (RAG)** component that provides nutrition-related question answering based on a selected diabetes nutrition reference. The RAG pipeline combines retrieval, semantic matching, reranking, and grounded generation to provide evidence-based responses rather than relying solely on an unrestricted generative model.

Therefore, this project proposes **Gluko Guide**, an intelligent web-based diabetes risk assessment and decision-support platform that combines Machine Learning, Explainable AI, Optical Character Recognition (OCR), and Retrieval-Augmented Generation. The system uses a trained **XGBoost model** to generate diabetes-related predictions, SHAP to explain the model's decisions, OCR to extract relevant laboratory values from uploaded medical reports, and a RAG-based question-answering component to provide grounded nutritional guidance.

The system is implemented as an interactive web platform that supports patient data entry, laboratory report processing, prediction visualization, explanation of prediction results, nutrition-related question answering, and a dedicated doctor dashboard for reviewing previously assessed patients.

1.2 Objectives

The main objective of this project is to design and develop an intelligent web-based healthcare system for diabetes prediction and personalized risk analysis using Machine Learning and Explainable AI techniques.

The specific objectives of the project are as follows:

1. To collect, preprocess, and analyze diabetes healthcare data.
2. To perform Exploratory Data Analysis (EDA) to identify important patterns and relationships within the dataset.
3. To develop and compare multiple Machine Learning models for diabetes prediction.
4. To evaluate model performance using different evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.
5. To integrate Explainable AI techniques such as SHAP for interpreting prediction results.
6. To identify the most influential healthcare and lifestyle factors affecting diabetes risk.

7. To generate personalized healthcare recommendations based on user data.
8. To implement a lifestyle simulation feature that estimates how behavioral changes may affect diabetes risk.
9. To design and develop an interactive web application for healthcare risk analysis.
10. To create a user-friendly healthcare decision support system that enhances awareness and preventive healthcare.

1.3 Problem Statement

Despite significant advancements in Machine Learning and healthcare analytics, several limitations remain in existing diabetes prediction systems.

First, many existing systems focus primarily on predictive performance without providing sufficient explanations for their predictions. A highly accurate prediction that cannot be interpreted may reduce user understanding and trust, particularly in healthcare applications.

Second, prediction alone does not provide users with evidence-based information about diabetes management and nutrition. After receiving a prediction, users may have additional questions that require reliable and clinically grounded answers.

Third, many research-based prediction models remain isolated experimental systems and are not implemented as practical interactive platforms. This limits their accessibility and reduces their potential usefulness for patients and healthcare professionals.

Another limitation is the dependence on manual data entry. Laboratory reports commonly contain important measurements required for diabetes assessment, yet manually transferring these values into a digital system can be time-consuming and may introduce transcription errors.

Furthermore, healthcare professionals require a mechanism to review previously assessed patients and identify individuals who may require additional attention. A prediction model without an appropriate monitoring interface provides limited support for practical clinical workflows.

Therefore, there is a need for an integrated system capable of:

- Predicting diabetes-related risk using clinical and lifestyle information.
- Explaining the factors contributing to model predictions.
- Extracting relevant laboratory values from uploaded medical reports.
- Providing evidence-grounded nutritional guidance.
- Presenting patient assessment results through an accessible web interface.

- Supporting healthcare professionals through a dedicated patient-monitoring dashboard.

This project addresses these requirements through the development of **Gluco Guide**, an integrated web-based diabetes risk assessment and decision-support platform.

1.4 Scope

This project focuses on the development of an intelligent web-based platform for diabetes risk assessment, prediction explanation, laboratory report processing, and evidence-grounded nutritional guidance.

The scope of the project includes:

- Using a publicly available diabetes healthcare dataset for model development and evaluation.
- Performing data preprocessing, encoding, feature engineering, and feature selection.
- Conducting Exploratory Data Analysis.
- Training and comparing multiple Machine Learning models.
- Selecting XGBoost as the final prediction model based on experimental performance.
- Integrating SHAP for model explainability.
- Developing a web-based patient assessment interface.
- Collecting the clinical and demographic variables required by the trained model.
- Automatically calculating BMI from height and weight.
- Validating user-entered clinical information.
- Processing uploaded laboratory reports using OCR.
- Extracting relevant laboratory values from medical reports.
- Allowing users to manually complete laboratory information that cannot be extracted automatically.
- Providing prediction results and model confidence through the web interface.
- Providing explanation components associated with the prediction.
- Integrating a RAG-based nutrition question-answering component grounded in a diabetes nutrition reference.
- Providing a dedicated doctor dashboard for reviewing patient assessments and statistical information.

The final predictive model uses the following primary patient variables:

- Age
- Gender
- Height
- Weight
- HbA1c
- Fasting Blood Glucose
- Postprandial Blood Glucose
- Total Cholesterol
- Systolic Blood Pressure
- Family History of Diabetes
- Physical Activity
- Insulin Level

The project does not replace professional medical diagnosis or treatment. The generated predictions and nutritional responses are intended to support awareness and decision-making rather than provide definitive medical diagnoses or treatment plans.

The current implementation does not include direct real-time integration with Continuous Glucose Monitoring (CGM) devices, hospital Electronic Health Record (EHR) systems, or clinical monitoring infrastructure. Such integrations are considered potential future extensions of the proposed platform.

1.5 Research Questions

This project aims to answer the following research questions:

1. How effectively can Machine Learning models classify diabetes-related conditions using the available clinical and lifestyle-related features?
2. Which Machine Learning model provides the best predictive performance for the proposed diabetes assessment task?
3. How can SHAP-based Explainable AI improve the transparency and interpretability of the final XGBoost predictions?
4. Which patient features contribute most significantly to the predictions generated by the selected model?

5. Can OCR reduce the effort required to manually enter laboratory measurements into the diabetes assessment system?
6. How can Retrieval-Augmented Generation provide grounded and source-based answers to diabetes nutrition-related questions?
7. How can Machine Learning prediction, explainability, OCR, and RAG be integrated into a single interactive web-based healthcare platform?
8. How can a dedicated doctor dashboard support the review and monitoring of previously assessed patients?

1.6 Proposed Solution Approach

The proposed solution consists of an integrated pipeline combining data analytics, Machine Learning, Explainable AI, OCR, RAG, and web technologies.

The first stage involves collecting and preprocessing the diabetes healthcare dataset. The data preparation process includes cleaning, encoding categorical variables, feature engineering, feature selection, and appropriate preprocessing operations.

The second stage involves Exploratory Data Analysis to investigate the distribution of the data and identify relationships between clinical variables and diabetes-related outcomes.

The third stage focuses on Machine Learning model development and comparative evaluation. Several classification algorithms are trained and evaluated using multiple performance metrics. Following the optimization and comparison stages, **XGBoost is selected as the final prediction model** due to its superior predictive performance and suitability for explainability and deployment.

The fourth stage integrates **SHAP** with the trained XGBoost model. SHAP provides both global feature importance and local explanations, allowing the system to identify the factors contributing to individual predictions.

The fifth stage introduces an **OCR-based laboratory report processing pipeline**. Users can upload a medical laboratory report, after which OCR extracts the text and the parsing component identifies relevant laboratory measurements. Successfully extracted values are transferred to the patient form, while missing values remain available for manual completion.

The sixth stage integrates the **Retrieval-Augmented Generation pipeline** into the web platform. Users can submit nutrition-related questions through a dedicated interface. The system retrieves relevant

information from the selected diabetes nutrition reference, reranks the retrieved content, and generates an answer grounded in the retrieved evidence.

The seventh stage focuses on web application implementation. The platform integrates the patient assessment interface, backend prediction services, OCR processing, prediction visualization, explanation components, and RAG-based nutrition question answering.

Finally, a **doctor dashboard** provides healthcare professionals with an overview of previously assessed patients, risk categories, and statistical summaries.

The overall concept can therefore be represented as:

Patient Data / Laboratory Report → Data Processing → XGBoost Prediction → SHAP Explanation

and, in parallel:

Nutrition Question → Hybrid Retrieval → Reranking → Grounded Generation → Evidence-Based Answer

These components are integrated within the same web-based platform to create a unified diabetes risk assessment and decision-support system.

1.7 Significance of the Project

The significance of this project lies in combining predictive analytics with interpretability, evidence-grounded information retrieval, and practical web-based deployment.

From a healthcare perspective, the system can support early risk assessment and improve awareness by presenting diabetes-related predictions together with explanations of the factors contributing to those predictions.

From a Machine Learning perspective, the project demonstrates the complete development cycle of a healthcare prediction system, including preprocessing, feature engineering, feature selection, model comparison, model selection, explainability, and deployment.

The integration of SHAP improves transparency by allowing users to understand which patient features influenced the prediction. This is particularly important in healthcare applications, where users and professionals may require more information than a simple predicted class.

The OCR component improves usability by reducing the need for users to manually transfer laboratory measurements from medical reports into the system.

The RAG component further extends the platform by providing nutrition-related answers grounded in a specific reference. This creates a distinction between the predictive component, which estimates the patient's diabetes-related status, and the information-support component, which provides evidence-grounded nutritional guidance.

The doctor dashboard also provides a practical mechanism for reviewing patient assessments and identifying high-risk cases.

Potential beneficiaries of the proposed system include:

- Individuals interested in diabetes risk assessment and health awareness.
- Healthcare professionals reviewing patient assessments.
- Healthcare researchers.
- Students and academic institutions.
- Developers of AI-based healthcare applications.
- Organizations interested in preventive healthcare technologies.

Overall, the project demonstrates how Machine Learning, Explainable AI, OCR, RAG, and web technologies can be combined to transform an experimental prediction model into an interactive healthcare decision-support prototype.

1.8 Project Outline

This report is organized into five main chapters as follows:

Chapter One: Introduction

This chapter presents the background of the project, problem statement, objectives, scope, research questions, proposed solution approach, significance, and overall structure of the project.

Chapter Two: Literature Review

This chapter reviews previous research related to diabetes prediction, Machine Learning, Explainable AI, OCR-based healthcare systems, and Retrieval-Augmented Generation. It discusses existing approaches, datasets, algorithms, performance measures, and limitations of previous studies.

Chapter Three: Methodology

This chapter presents the methodology used to develop the proposed system. It covers dataset preparation, Exploratory Data Analysis, feature engineering, feature selection, preprocessing, Machine

Learning model development, comparative evaluation, PCA experiments, XGBoost selection, SHAP explainability, and the RAG pipeline.

Chapter Four: System Design and Implementation

This chapter describes the implementation of the proposed web-based platform. It presents the system architecture, frontend and backend components, patient data collection, clinical measurements, medical history, XGBoost integration, OCR-based laboratory report processing, prediction visualization, SHAP-based explanation, RAG integration, doctor dashboard, high-risk patient monitoring, frontend–backend communication, validation and error handling, user interface design, and the overall system workflow.

Chapter Five: Conclusion and Future Work

This chapter summarizes the main findings and contributions of the project. It discusses the selected XGBoost model, explainability through SHAP, OCR-based data extraction, RAG-based nutritional guidance, and web-based deployment. It also presents the limitations of the current prototype and proposed future extensions, including integration with Continuous Glucose Monitoring devices, healthcare clinics, healthcare professionals, longitudinal patient monitoring, Electronic Health Records, and expansion of the RAG knowledge base.

CHAPTER 2

BACKGROUND & LITERATURE REVIEW

2.1 Background

Diabetes mellitus represents one of the most significant global public health challenges due to its increasing prevalence, chronic complications, and substantial economic burden on healthcare systems. Early detection and risk prediction are essential for reducing long-term complications such as cardiovascular disease, kidney failure, neuropathy, and retinopathy. Consequently, predictive healthcare systems based on Artificial Intelligence (AI) and Machine Learning (ML) have emerged as promising approaches for improving diabetes diagnosis, risk stratification, and clinical decision support [1], [2].

The rapid advancement of Machine Learning in healthcare has transformed traditional diagnostic methodologies by enabling data-driven prediction and automated decision-making. Unlike conventional statistical approaches that rely primarily on predefined assumptions, ML models can identify hidden nonlinear relationships among clinical, behavioral, demographic, and physiological variables.

This capability has significantly improved predictive analytics in healthcare applications, particularly in chronic disease prediction systems where complex interactions among risk factors exist [1], [3], [4].

In diabetes prediction research, ML-based clinical decision support systems have been widely adopted to assist healthcare professionals in identifying high-risk individuals before disease progression occurs. Predictive analytics enables healthcare systems to move from reactive treatment models toward preventive and personalized healthcare paradigms. Through analysis of patient histories, laboratory findings, lifestyle behaviors, and comorbidities, predictive models can support early intervention strategies and improve healthcare resource allocation [1], [4], [5], [6].

The importance of diabetes prediction research has further increased due to the availability of large healthcare datasets such as the Pima Indians Diabetes Dataset (PIDD), Behavioral Risk Factor Surveillance System (BRFSS), and various clinical cohort datasets.

These datasets have enabled researchers to evaluate both traditional machine learning algorithms and advanced ensemble or deep learning techniques under diverse healthcare conditions. However, despite substantial improvements in predictive performance, many existing studies remain limited by issues related to data imbalance, poor generalizability, lack of interpretability, and limited deployment into practical healthcare environments [1], [2], [7], [8], [9].

More recently, Explainable Artificial Intelligence (XAI) has become a critical research direction in healthcare prediction systems. Traditional black-box models often achieve high predictive accuracy but fail to provide understandable reasoning behind predictions, limiting clinical trust and adoption. Explainability techniques such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations), and Partial Dependence Plots (PDP) have therefore been increasingly integrated into diabetes prediction frameworks to improve transparency and support clinician interpretation [5], [6], [10], [11], [12].

Simultaneously, the growing emphasis on personalized healthcare systems has shifted research attention toward individualized recommendations and lifestyle-aware predictive models. Rather than producing only binary prediction outputs, modern healthcare AI systems increasingly aim to provide interpretable risk explanations, behavioral insights, and personalized preventive guidance tailored to each patient's clinical and lifestyle characteristics. Consequently, contemporary diabetes prediction research now combines predictive analytics, explainability, and interactive healthcare delivery systems to improve both predictive performance and practical healthcare usability [6], [9], [11].

2.2 Datasets Used in Previous Studies

The effectiveness of diabetes prediction systems is strongly influenced by dataset quality, feature diversity, population representation, and preprocessing methodology. Across the reviewed literature, researchers relied primarily on four categories of datasets: the Pima Indians Diabetes Dataset (PIDD), BRFSS datasets, lifestyle-oriented healthcare datasets, and clinical cohort datasets. Each dataset category presents distinct advantages and limitations that directly affect model generalizability, prediction accuracy, and healthcare applicability [1], [7], [8], [9].

The Pima Indians Diabetes Dataset remains the most widely used benchmark dataset in diabetes prediction research. Studies conducted by Butt et al.[1], El Sherbiny et al.[13], Ashisha et al.[14], Iparraguirre-Villanueva et al.[2], and several others utilized the PIDD due to its accessibility and standardized structure. The dataset contains 768 instances and nine clinical attributes including plasma

glucose concentration, BMI, insulin level, skin thickness, pregnancies, blood pressure, diabetes pedigree function, and age. Its widespread adoption has enabled consistent benchmarking across machine learning algorithms.

However, the dataset suffers from several major limitations. First, the population is restricted to female participants of Pima heritage above the age of 21, substantially limiting demographic diversity and external generalizability. Second, the dataset contains physiologically implausible zero values in multiple clinical features, creating preprocessing challenges that can artificially inflate predictive performance when not handled appropriately. Third, the dataset exhibits moderate class imbalance, with approximately 65% non-diabetic and 35% diabetic cases.

Several studies attempted to address these limitations through preprocessing and balancing techniques. Ashisha et al.[14] applied median imputation to replace unrealistic zero values and used feature normalization and hyperparameter optimization to improve data quality. Similarly, Iparraguirre-Villanueva et al.[2] evaluated the impact of SMOTE and PCA on classification performance, demonstrating that class balancing substantially improved minority-class recall while PCA significantly degraded predictive accuracy. These findings suggest that preprocessing strategy selection is highly dataset-dependent and critically influences healthcare prediction outcomes [15].

In contrast to the relatively small PIDD dataset, the Behavioral Risk Factor Surveillance System (BRFSS) datasets provide substantially larger and more diverse population samples. Studies by Allani[6], Khokhar et al.[11], Nguyen and Zhang[8], Ullah et al.[16], and Weaver[7] employed BRFSS datasets containing more than 250,000 survey responses collected by the CDC. Unlike PIDD, BRFSS datasets incorporate extensive behavioral, demographic, socioeconomic, and health-history variables including smoking behavior, alcohol consumption, physical activity, mental health, cholesterol status, hypertension, cardiovascular disease history, education level, and income.

The large sample size and demographic diversity of BRFSS datasets significantly improve statistical robustness and population-level generalizability. Furthermore, lifestyle-oriented variables enable researchers to investigate behavioral risk factors associated with diabetes development rather than relying solely on laboratory biomarkers. However, BRFSS datasets also introduce important limitations. Since most variables are self-reported, the datasets are susceptible to recall bias, reporting bias, and inconsistencies in survey responses[7].

In addition, class imbalance remains severe due to the relatively lower prevalence of diabetic cases compared to non-diabetic participants. Consequently, many studies applied oversampling approaches

such as SMOTE, SMOTE-Tomek, undersampling, or hybrid balancing strategies to improve minority-class representation [15].

Lifestyle-oriented healthcare datasets were particularly important in studies emphasizing preventive healthcare and accessible diabetes screening. Nguyen and Zhang[8] demonstrated that lifestyle and behavioral variables alone could support meaningful diabetes prediction, although predictive performance remained lower than models incorporating direct clinical biomarkers. Similarly, Weaver’s work on BRFSS 2022 highlighted the predictive value of alcohol intake, smoking behavior, and mental health indicators[7].

These findings collectively suggest that behavioral datasets can contribute significantly to population-level risk assessment systems, particularly in low-resource healthcare environments where laboratory testing may be limited.

Clinical cohort datasets provided another important category of healthcare data. Rafie et al[9]. utilized the Dena Cohort dataset from Iran containing 3,203 participants and a comprehensive set of biochemical, lifestyle, and comorbidity-related features. Similarly, El Sherbiny et al[13]. developed a diabetes prediction framework using an Egyptian clinical dataset that incorporated HbA1c measurements, lipid profiles, dietary behavior, smoking status, and physical activity indicators. Haque et al[17]. also incorporated a private Bangladeshi hospital dataset to complement the PIDD benchmark.

Compared with publicly available benchmark datasets, clinical cohort datasets often provide richer clinical representations and improved regional applicability. They also allow inclusion of population-specific epidemiological characteristics that may not exist in international datasets. However, these datasets are frequently limited by smaller sample sizes, restricted geographic representation, and reduced reproducibility due to limited public accessibility [9], [13].

Across the reviewed studies, dataset selection significantly influenced algorithmic performance and healthcare applicability. Smaller datasets such as PIDD generally favored traditional machine learning models and risked overfitting under deep learning architectures. Conversely, large-scale datasets such as BRFSS enabled ensemble learning methods to achieve stronger generalizability and more robust feature discovery. Moreover, studies incorporating both lifestyle and clinical features consistently demonstrated improved predictive capability compared to models relying exclusively on a single feature category [3], [4].

Overall, the literature demonstrates that dataset quality, diversity, balancing strategy, and feature composition are among the most influential determinants of diabetes prediction system effectiveness. The persistent limitations related to demographic representation, class imbalance, and dataset

generalizability continue to motivate the development of more comprehensive and diverse healthcare prediction frameworks.

2.3 Methodologies Used in Previous Studies

2.3.1 Machine Learning in Healthcare

Machine Learning has become one of the most influential technological advancements in modern healthcare due to its capability to extract predictive patterns from complex clinical and behavioral data. In healthcare prediction systems, ML algorithms are increasingly used for disease diagnosis, patient stratification, treatment optimization, risk forecasting, and clinical decision support. Diabetes prediction has emerged as one of the most extensively studied healthcare applications because of the disease's multifactorial nature and the availability of structured healthcare datasets [3], [4], [18].

The reviewed literature demonstrates a progressive evolution from simple statistical classification toward advanced AI-driven healthcare analytics. Early studies focused primarily on improving classification accuracy using traditional supervised learning algorithms. More recent studies integrate ensemble learning, deep learning, explainable AI, imbalance handling techniques, and interactive deployment frameworks to improve both predictive performance and clinical usability [3], [5], [6], [9], [18].

Predictive analytics in healthcare systems relies heavily on preprocessing quality, feature engineering, imbalance correction, and model interpretability[6]. Several studies emphasized that preprocessing strategies such as normalization, imputation, feature selection, and oversampling significantly influence predictive performance[14]. Elluri and Abinay[15], and Haque et al[17]. demonstrated that robust preprocessing pipelines substantially improve model stability and minority-class prediction accuracy.

In addition, healthcare ML systems increasingly emphasize interpretability due to the critical consequences of medical decision-making. Consequently, contemporary diabetes prediction research has shifted toward explainable, transparent, and clinically interpretable AI systems rather than purely accuracy-oriented black-box architectures [9], [10], [11], [12].

2.3.2 Traditional Machine Learning Models

Logistic Regression

Logistic Regression (LR) is one of the most widely used statistical classification techniques in healthcare prediction systems due to its simplicity, interpretability, and computational efficiency.

The model estimates the probability of class membership through a logistic function, making it particularly suitable for binary classification tasks such as diabetic versus non-diabetic prediction [4]. Several reviewed studies employed Logistic Regression either as a baseline classifier or as part of comparative evaluations. Nguyen and Zhang[8] reported that Logistic Regression achieved the highest performance among Decision Tree and KNN models on BRFSS lifestyle data, reaching approximately 75% accuracy. Similarly, El Sherbiny et al[13]. demonstrated competitive performance using Logistic Regression on an Egyptian clinical dataset, while Elluri and Abinay achieved 78.57% accuracy following preprocessing and normalization [15].

The primary strength of Logistic Regression lies in its interpretability. Model coefficients provide direct insight into feature contributions, making the algorithm relatively transparent compared to complex black-box models. Furthermore, Logistic Regression performs efficiently on smaller structured tabular datasets and requires relatively low computational resources [4], [18].

However, Logistic Regression exhibits several limitations in diabetes prediction tasks. The model assumes linear relationships between predictors and outcomes, reducing its ability to capture complex nonlinear healthcare interactions. Consequently, Logistic Regression generally underperforms compared to ensemble learning methods and advanced nonlinear classifiers. In large-scale healthcare datasets with heterogeneous feature interactions, the model's predictive capability often becomes insufficient for high-performance clinical applications.

Despite these limitations, Logistic Regression remains an important benchmark model in healthcare ML research due to its interpretability, stability, and clinical transparency [4], [18].

Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm designed to maximize the margin between classification boundaries through optimal hyperplane construction. SVM is particularly effective in high-dimensional healthcare datasets and nonlinear classification tasks through the use of kernel functions [2], [18].

Several reviewed studies demonstrated strong SVM performance in diabetes prediction. Haque et al[17]. reported that SVM combined with SMOTE achieved 95% accuracy and 91% F1-score, substantially outperforming the non-balanced version of the same model. Hoyos et al. also achieved approximately 99% accuracy using SVM following normalization and class balancing[19]. Weaver further demonstrated strong SVM performance on lifestyle-oriented BRFSS datasets [7].

The principal advantage of SVM lies in its robustness to high-dimensional feature spaces and its ability to model nonlinear relationships effectively. In healthcare prediction systems where complex interactions exist among demographic, physiological, and behavioral variables, SVM often achieves stronger classification boundaries than linear approaches [2], [18].

Nevertheless, SVM presents several practical limitations. Model training becomes computationally expensive for large-scale datasets such as BRFSS, and kernel selection significantly influences predictive performance. Furthermore, SVM models inherently lack interpretability, limiting direct clinical understanding of prediction rationale unless complemented by XAI techniques such as SHAP or LIME [10].

Compared to Logistic Regression, SVM generally achieves stronger predictive performance but sacrifices interpretability and computational simplicity [4], [18].

Decision Trees and Random Forest

Decision Trees represent one of the most interpretable machine learning approaches used in healthcare analytics. The algorithm recursively partitions feature spaces based on information gain or impurity reduction to generate rule-based classification structures. Due to their transparency and human-readable structure, Decision Trees are commonly applied in medical decision support systems [4], [18].

However, standalone Decision Trees often suffer from instability and overfitting.

Several reviewed studies demonstrated relatively weak Decision Tree performance compared to ensemble methods. Iparraguirre-Villanueva et al. reported only 63% accuracy using Decision Trees on the PIDD dataset [2].

To overcome these limitations, Random Forest ensemble learning methods were extensively adopted. Random Forest constructs multiple decision trees using bootstrap aggregation and random feature selection, substantially improving stability, generalization, and predictive accuracy. Panda and Mahanta[10], El Sherbiny et al.[13], Ashisha et al.[14], and Elluri and Abinay[15] all reported strong Random Forest performance across different diabetes datasets.

The major strength of Random Forest lies in its balance between predictive accuracy and interpretability. Feature importance analysis enables identification of clinically relevant predictors such as glucose level, BMI, and age. Moreover, Random Forest models are relatively robust to noise, missing values, and overfitting compared to single-tree architectures [10], [13].

Nevertheless, Random Forest models remain less interpretable than simple statistical models and often require explainability frameworks to support clinical trust. Consequently, many recent studies integrated SHAP and LIME with Random Forest classifiers to improve interpretability [9], [10], [11], [12].

2.3.3 Advanced Models (Ensemble Learning and Deep Learning)

The reviewed literature demonstrates a clear transition from traditional machine learning toward advanced ensemble learning and deep learning methodologies aimed at improving predictive performance, robustness, and healthcare applicability.

Among ensemble methods, XGBoost emerged as one of the most consistently high-performing algorithms across multiple studies. Rafie et al.[9], Khokhar et al.[11], Hoyos et al.[19], and Islam et al.[12] all demonstrated superior performance using XGBoost-based architectures. The algorithm's gradient boosting mechanism enables sequential error correction, regularization, and efficient handling of heterogeneous healthcare features.

Gradient Boosting and AdaBoost similarly demonstrated strong predictive capability by iteratively combining weak learners into highly optimized ensemble structures. Khokhar et al.[11] and Islam et al.[12] reported strong ensemble performance under imbalanced healthcare conditions following hyperparameter optimization.

Bagging-based approaches such as Random Forest also remained highly competitive due to their stability and resistance to overfitting. Hybrid and ensemble systems proposed by Elluri and Abinay[15] further improved predictive performance by integrating probability outputs from multiple classifiers.

Deep learning approaches including ANN, CNN, and LSTM were primarily investigated in studies focusing on complex healthcare modalities such as retinal imaging, glucose forecasting, and temporal healthcare data. Sharma and Shah's[4] review highlighted that deep learning architectures often achieved accuracies exceeding 95% in several studies. CNN models demonstrated exceptional performance in image-based diabetic retinopathy detection, while LSTM architectures showed strong capability for temporal glucose prediction [18].

However, recent literature critically observed that deep learning models frequently underperform gradient-boosted ensemble methods on small structured tabular datasets such as PIDD. This finding is particularly important because most diabetes screening systems rely primarily on structured clinical and lifestyle data rather than high-dimensional imaging or longitudinal temporal signals [4], [18].

Although deep learning approaches offer strong representation-learning capability, they introduce significant computational complexity, overfitting risks, and interpretability challenges. Consequently, many recent healthcare studies prioritize ensemble learning approaches that provide competitive performance while maintaining relatively lower computational cost and greater explainability compatibility [9], [11], [19].

2.3.4 Explainable AI in Healthcare

Explainable Artificial Intelligence has become one of the most critical research areas in healthcare AI due to the need for transparency, trust, accountability, and clinical interpretability. In medical decision support systems, black-box predictions without understandable reasoning are often insufficient for real-world healthcare adoption [9], [10], [11], [12].

Among the reviewed studies, SHAP emerged as the most widely adopted explainability technique. Islam et al.[12], Khokhar et al.[11], Rafie et al.[9], Panda and Mahanta[10], and Hoyos et al.[19] all integrated SHAP-based explanation frameworks into their prediction systems. SHAP provides both global and local interpretability by quantifying each feature's contribution relative to a baseline prediction.

Islam et al.[12] presented the most comprehensive explainability framework by combining SHAP, LIME, and Partial Dependence Plots within a unified architecture. SHAP was used for feature importance analysis, PDP visualized feature-risk relationships, and LIME generated local human-readable explanations. Similarly, Khokhar et al. integrated SHAP, LIME, PDP, Anchors, Counterfactual Explanations, and Explainable Boosting Machines to improve healthcare interpretability [11].

LIME was particularly valuable for generating local explanation rules for individual predictions. Studies by Panda and Mahanta[10] and Islam et al.[12] demonstrated that LIME explanations improve transparency for patient-specific predictions. However, several studies concluded that SHAP generally provides more stable and mathematically consistent explanations than LIME for global interpretation tasks [9], [11].

Partial Dependency Plots were primarily used to visualize nonlinear relationships between risk factors and prediction outcomes. Islam et al.[12] demonstrated that PDPs effectively reveal how variables such as BMI, glucose level, and age influence diabetes risk probabilities across populations [9], [11].

Despite substantial explainability progress, several limitations remain evident across the literature. Many XAI outputs are designed primarily for technical researchers rather than healthcare practitioners or patients. Furthermore, numerous studies fail to translate explainability outputs into actionable healthcare

recommendations. Consequently, contemporary research increasingly focuses on patient-friendly explainability frameworks integrated into real-time healthcare applications [9], [10], [11].

Overall, the literature strongly supports the integration of XAI mechanisms into diabetes prediction systems to improve transparency, accountability, and clinical trust [12], [11], [10], [9]

2.3.5 Lifestyle and Risk Factor Analysis Studies

A growing body of diabetes prediction research emphasizes the importance of lifestyle and behavioral factors in healthcare analytics. Unlike purely clinical prediction systems based exclusively on laboratory biomarkers, lifestyle-aware prediction frameworks attempt to model modifiable risk behaviors associated with diabetes development [4], [9], [18], [19].

Across the reviewed studies, BMI consistently emerged as one of the strongest diabetes predictors. SHAP analyses conducted by Khokhar et al.[11], Panda and Mahanta[10], Rafie et al.[9], and Hoyos et al.[19] repeatedly identified BMI among the most influential risk factors .

Smoking behavior, alcohol consumption, physical activity, and sleep quality were also strongly associated with diabetes risk in BRFSS-based studies. Alzboon et al.[18] demonstrated that lifestyle-oriented variables can support meaningful diabetes prediction even without laboratory biomarkers. Similarly, Hoyos et al.[19] identified smoking, alcohol intake, and physical health indicators as important predictive variables.

Studies incorporating socioeconomic and behavioral variables demonstrated that healthcare prediction extends beyond physiological measurements alone. Factors such as income, education, cardiovascular disease history, hypertension, and dietary behavior significantly influenced diabetes risk profiles across multiple populations [4], [18], [19].

The integration of lifestyle variables provides several important advantages. First, behavioral variables support preventive healthcare by identifying modifiable risk factors. Second, lifestyle-aware systems can improve accessibility in environments where laboratory testing is unavailable or expensive. Third, personalized recommendations become more meaningful when grounded in behavioral analysis rather than solely biochemical measurements [11], [18], [19].

However, lifestyle-based datasets also introduce limitations related to self-reported data reliability, recall bias, and demographic variability. Consequently, studies combining both clinical biomarkers and lifestyle variables generally achieved stronger predictive performance and broader healthcare applicability.

The literature therefore increasingly supports hybrid healthcare prediction frameworks integrating clinical, demographic, behavioral, and socioeconomic features within unified predictive systems. [4], [11], [18], [19].

2.4 Comparative Analysis of Existing Work

The reviewed literature demonstrates substantial methodological diversity in diabetes prediction research, particularly regarding datasets, preprocessing strategies, algorithm selection, explainability integration, and deployment capability.

Studies relying on the Pima Indians Diabetes Dataset generally focused on benchmarking traditional machine learning algorithms due to the dataset's small size and structured clinical attributes. While these studies enabled standardized algorithm comparison, their generalizability remained limited due to demographic restriction and insufficient feature diversity. In contrast, BRFSS-based studies incorporated significantly larger population samples and broader lifestyle-related features, substantially improving population-level applicability and behavioral risk analysis capability [4], [18], [19].

Regarding preprocessing methodology, several studies demonstrated that preprocessing quality strongly influences predictive performance. Recent reviews highlighted the importance of handling physiologically implausible values through robust imputation strategies, while studies by Alzboon et al. [18] and Islam et al. [12] demonstrated that imbalance handling methods such as SMOTE substantially improve minority-class sensitivity. Conversely, comparative analyses showed that aggressive dimensionality reduction techniques like PCA may degrade predictive performance on smaller structured healthcare datasets [4].

Traditional machine learning algorithms such as Logistic Regression, KNN, SVM, and Decision Trees generally achieved moderate performance ranging between 70% and 85% accuracy across most studies. Logistic Regression consistently provided strong interpretability but weaker nonlinear modeling capability. SVM demonstrated stronger classification boundaries but introduced computational and interpretability limitations [4], [18], [19].

Ensemble learning methods consistently outperformed traditional classifiers across large-scale and imbalanced healthcare datasets. XGBoost, LightGBM, Gradient Boosting, and Random Forest repeatedly achieved superior accuracy, ROC-AUC, and F1-scores. Rafie et al. [9] reported one of the

strongest performances using XGBoost with 96.07% accuracy and 99.29% AUC, while Khokhar et al. achieved 92.80% accuracy using a soft-voting ensemble architecture [11].

Deep learning approaches demonstrated strong performance primarily in imaging and temporal healthcare applications. However, comprehensive reviews such as Sharma and Shah[4] concluded that ensemble learning often outperforms deep learning on structured tabular healthcare datasets while requiring lower computational complexity [10], [18].

Explainable AI integration represented one of the most important distinctions between contemporary and earlier healthcare ML systems. Earlier studies primarily focused on maximizing predictive accuracy without interpretability mechanisms. More recent works by Islam et al.[12], Khokhar et al.[11], Rafie et al.[9], and Panda and Mahanta[10] demonstrated a significant transition toward transparent healthcare AI systems integrating SHAP, LIME, PDP, Anchors, and Counterfactual Explanations [10].

Despite these advancements, many studies remained limited to research-oriented analytical environments without practical healthcare deployment. Only a small number of studies implemented interactive web-based healthcare systems integrating prediction, explainability, and personalized recommendations. Even among deployed systems, clinical validation, accessibility compliance, and patient-oriented explanation design often remained insufficient [9], [10], [11].

Another critical observation across the literature concerns feature integration strategy. Studies relying exclusively on laboratory biomarkers generally achieved stronger predictive performance but weaker accessibility. Conversely, lifestyle-based systems improved accessibility but often produced lower predictive accuracy. Hybrid systems combining clinical and lifestyle features consistently demonstrated the strongest balance between predictive capability and healthcare applicability.

Overall, the comparative analysis demonstrates that modern diabetes prediction research increasingly prioritizes integrated healthcare systems combining robust preprocessing, ensemble learning, explainable AI, personalized healthcare recommendations, and real-time deployment capabilities [9], [10], [11], [12], [19].

Table (2.1) Comparative Analysis of Existing Work

Author / Year	Dataset	Algorithm(s)	Best Accuracy	Limitation
Zhu et al. (2021)	PIDD, CGM, Retinopathy datasets	LSTM, CNN, RNN, Autoencoder	AUC > 0.97 (CNN)	No external validation; black-box DL; no deployment
Butt et al. (2021)	PIDD (768 records)	MLP, RF, LR (class.); LSTM, MA, LR (pred.)	MLP: 86.08%; LSTM: 87.26%	Single dataset; no XAI; IoT module hypothetical only
Sharma & Shah (2021)	Review — PIDD, EHR, Retinal, Wearable	SVM, LR, DT, ANN, LSTM, CNN (survey)	75–99% across reviewed studies	Review only; no new model; XAI gap identified not solved
Belal et al. (2022)	PIDD (768 records)	RF, SVM, DT, k-NN, NB, LR, GBM	~79% (RF)	Single dataset; no imbalance handling; no XAI
Kibria et al. (2022)	PIDD (768 records)	Soft-Voting Ensemble + SHAP + LIME	82.3%, AUC=0.89	Single dataset; XAI not in patient-accessible interface
Ullah et al. (2022)	UCI Early-Stage Diabetes (520 records)	RF, DT, SVM, k-NN, NB	97.4% (RF)	Small symptom-only dataset; overfitting risk; no XAI
Iparraguirre-Villanueva et al. (2023)	PIDD + supplemental clinical records	GBM, RF, SVM, MLP, LR, DT, k-NN (7)	82.7% (GBM)	No XAI; no web interface; no prospective validation
Panda & Mahanta (2023)	PIDD (768 records)	RF + SHAP + LIME + SHAP Interaction Values	~79% (RF)	Single small dataset; no imbalance handling; no deployment
El-Sappagh et al. (2023)	Clinical registry + PIDD	XGBoost, RF, GBM + SHAP (global + local)	91.3%, AUC=0.94 (XGBoost)	No clinical deployment; no personalised recommendations

Author / Year	Dataset	Algorithm(s)	Best Accuracy	Limitation
Hoyos et al. (2024)	Small clinical dataset (sociodemographic)	FCM, ANN, SVM, XGBoost + Chi-square	High classification efficacy	Small dataset; FCM complex; no deployment; no recommendations
Mansouri et al. (2024)	PIDD + expanded clinical dataset	RF, GBM, SVM, LDA, LR, Bagging (10 total)	82.4%, AUC=0.89 (RF)	Offline batch only; no XAI; no user interface
Kavitha et al. (2024)	PIDD (768 records)	XGBoost, RF, SVM, AdaBoost, DT, k-NN, NB	84.7% (XGBoost+SMOTE)	Single dataset; no XAI; no deployment interface
Weaver (2024)	BRFSS 2022 (national CDC survey)	LR, RF, Gradient Boosting, MLP	78%, AUC=0.83 (GBM)	No deployment; self-reported data; no formal XAI
El Sherbiny et al. (2025)	Egyptian clinical dataset	RF, SVM, LR, DT, Naïve Bayes	83.1%, AUC=0.91 (RF)	No XAI; no deployment; retrospective validation only
Khokhar et al. (2025)	BRFSS Health Indicators (Kaggle/CDC)	Ensemble + SHAP (XAI)	92.50%, AUC=0.975	Preprint; no deployment; no personalised recommendations
Nguyen & Zhang (2025)	BRFSS lifestyle-curated features	LightGBM, XGBoost, RF, MLP, Ensemble (9)	80.3%, AUC=0.845 (LightGBM)	No XAI; no deployment; cross-sectional only
XGBoost & XAI Study (2025)	Dena Cohort, Iran (3,203 records)	XGBoost, CatBoost + SHAP (global + local)	96.07%, AUC=0.993 (XGBoost)	No web deployment; geographically restricted; no recommendations
Allani (2025)	CDC BRFSS 2015 (national survey)	LightGBM, XGBoost, RF, LR, KNN, NN + SHAP + LIME	Best recall (LightGBM+under.)	2015 data; preprint; clinical validation of recommendations pending

Author / Year	Dataset	Algorithm(s)	Best Accuracy	Limitation
Islam et al. (2025)	Imbalanced clinical datasets (JnU)	GBM, XGBoost, RF + SHAP + PDP + LIME	Competitive AUC & F1	No patient-facing deployment; PDP limited to marginal effects

2.5 Research Gap Analysis:

Despite significant advancements in Machine Learning for diabetes prediction, several important limitations remain unresolved across the existing literature. Most current studies focus primarily on improving predictive accuracy while giving limited attention to interpretability, user interaction, and practical healthcare usability. Although high-performance models have been widely reported, many systems still function mainly as research-oriented prediction frameworks rather than comprehensive healthcare support platforms.

One major limitation involves the lack of integrated systems that combine prediction, explainability, personalized recommendations, and interactive deployment within a unified architecture. In many existing studies, users receive only a prediction result without understanding the factors influencing the prediction or the actions that may help reduce their diabetes risk. Consequently, many systems remain descriptive rather than actionable.

Another important gap concerns Explainable Artificial Intelligence (XAI). While some recent studies have incorporated SHAP, LIME, and other explainability techniques, explanation outputs are often highly technical and difficult for non-specialist users to understand. As a result, the transparency and practical usability of many prediction systems remain limited.

Similarly, personalized recommendation generation remains insufficiently explored in the literature. Although several studies identify influential healthcare and lifestyle factors associated with diabetes, very few systems provide individualized preventive guidance directly linked to the prediction rationale. This limits the ability of users to translate prediction results into meaningful behavioral changes.

Lifestyle simulation and counterfactual risk analysis also remain largely absent from existing diabetes prediction systems. Most reviewed frameworks do not allow users to explore how modifying variables such as BMI, smoking behavior, physical activity, or sleep habits may influence predicted diabetes risk. The absence of such interactive functionality reduces the preventive healthcare value of current systems.

In addition, most previous studies remain limited to offline experimental environments without real-world interactive deployment. Many prediction models are evaluated only within notebooks or research settings and are not implemented as accessible web-based healthcare applications suitable for ordinary users.

Another challenge relates to feature integration and dataset diversity. Numerous studies rely exclusively on either clinical biomarkers or lifestyle-related variables rather than combining both categories within a unified predictive framework. Furthermore, heavy dependence on benchmark datasets such as the Pima Indians Diabetes Dataset continues to limit demographic diversity and generalizability.

The present project addresses these identified research gaps through the development of an explainable and interactive healthcare decision support system for diabetes prediction and personalized risk analysis. The proposed framework integrates Machine Learning prediction models, SHAP-based explainability, personalized recommendation generation, lifestyle simulation features, and interactive web-based deployment within a single unified platform.

Unlike many existing systems, the proposed project aims not only to predict diabetes risk but also to improve transparency, accessibility, healthcare awareness, and actionability for end users. By combining prediction, interpretation, behavioral guidance, and counterfactual lifestyle analysis within an accessible web application, the project attempts to bridge the gap between research-oriented prediction models and practical patient-centered healthcare support systems.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter presents the methodology adopted for developing the proposed diabetes prediction and personalized risk-analysis system. Building on the research gaps identified in Chapter 2 namely the limited interpretability, limited generalizability and lack of practical deployment characteristic of many existing diabetes prediction studies this chapter details the complete technical pipeline through which the proposed framework was constructed, from raw data acquisition to model selection and knowledge-grounded question answering.

The methodology begins with the Diabetes Healthcare Dataset, a large-scale collection of 100,000 patient records spanning 31 demographic, lifestyle, medical-history, and clinical features, obtained from Kaggle (Section 3.2). Following an initial data preprocessing stage encompassing missing-value and duplicate detection, categorical encoding, and feature scaling the dataset was subjected to an in-depth Exploratory Data Analysis (Section 3.3), examining data quality, class distribution, demographic patterns, and correlations among numerical and categorical variables to inform subsequent modeling decisions.

These insights guided a structured feature engineering pipeline (Section 3.4), in which the data were stratified into training, validation, and test partitions, potentially leakage-inducing variables were excluded, a clinically motivated interaction feature (the postprandial-to-fasting glucose ratio) was derived, and low-information attributes were removed based on Mutual Information analysis. To address the mild class imbalance present in the target variable and ensure robust learning, SMOTE was applied exclusively to the training partition prior to model development (Section 3.5).

Five machine learning algorithms—Support Vector Machine, Random Forest, XGBoost, LightGBM, and Multi-Layer Perceptron—were implemented and systematically compared to identify the most effective predictive approach, with hyperparameter optimization performed via GridSearchCV and stratified cross-validation (Section 3.6). In addition, a separate Neural Network architecture was developed using TensorFlow to evaluate the performance of a feedforward deep learning approach for multi-class diabetes-stage prediction. Model explainability was incorporated using SHAP to quantify

the contribution of individual features to model predictions, while MLflow was used to systematically track experiments, hyperparameters, evaluation metrics, and model artifacts.

Finally, this chapter introduces the Adapted Retrieval-Augmented Generation (RAG) pipeline (Section 3.7), a complementary component designed to answer clinical nutrition queries by grounding generated responses in a dedicated reference text on the nutritional management of diabetes. This pipeline extends the system beyond numerical risk prediction, enabling it to provide evidence-based, source-cited responses to clinical nutrition questions.

Taken together, the stages described in this chapter—dataset preparation, preprocessing, exploratory analysis, feature engineering, class-imbalance handling, comparative machine learning development, neural network modeling, SHAP-based explainability, experiment tracking, and retrieval-augmented question answering—constitute an integrated methodology aimed at developing a diabetes prediction system that is accurate, interpretable, and practically deployable.

3.2 Dataset

3.2.1 Description

The dataset used in this study is the Diabetes Healthcare Dataset, obtained from Kaggle, and was specifically selected due to its comprehensive representation of factors associated with diabetes risk and diagnosis. The dataset contains 100,000 patient records and 31 features, providing a large-scale and diverse healthcare dataset suitable for machine learning applications.

The dataset incorporates multiple categories of information, including demographic characteristics, lifestyle behaviors, medical history indicators, anthropometric measurements, vital signs, laboratory test results, and diabetes-related clinical assessments. Demographic variables include age, gender, ethnicity, education level, income level, and employment status. Lifestyle-related attributes describe patients' smoking habits, alcohol consumption, physical activity levels, dietary quality, sleep duration, and daily screen time. These lifestyle factors are associated with the development and progression of diabetes.

In addition, the dataset contains medical history variables such as family history of diabetes, hypertension history, and cardiovascular disease history. Clinical and physiological measurements include Body Mass Index (BMI), waist-to-hip ratio, systolic and diastolic blood pressure, resting heart rate, cholesterol levels (total, HDL, and LDL), triglycerides, fasting glucose, postprandial glucose, insulin level, and HbA1c. These measurements are widely used in clinical practice for assessing

metabolic health and identifying diabetes-related conditions.

The target variable used in this study is **diagnosed_diabetes**, formulated as a **multiclass classification problem** with five outcome categories: **No Diabetes, Pre-Diabetes, Type 1 Diabetes, Type 2 Diabetes, and Gestational Diabetes**. Other diabetes-related variables were reviewed during preprocessing to identify potential target proxies and sources of data leakage. In particular, **diabetes_risk_score** and **Diagnosis_encoded** were excluded from the predictive feature set before model training to prevent information leakage and ensure a realistic evaluation of model performance.

The availability of both clinical and lifestyle-related attributes makes this dataset particularly valuable for developing predictive models capable of identifying diabetes risk factors and supporting early diagnosis. Furthermore, the large sample size enhances the reliability of statistical analysis and improves the generalization capability of machine learning models. Therefore, the dataset represents an appropriate foundation for building an accurate, interpretable, and practical diabetes prediction system.

Feature Categories

Table (3.1) Feature Categories and Included Variables

Category	Features Included
Demographic Features	age, gender, ethnicity, education_level, income_level, employment_status
Lifestyle Features	smoking_status, alcohol_consumption_per_week, physical_activity_minutes_per_week, diet_score, sleep_hours_per_day, screen_time_hours_per_day
Medical History Features	family_history_diabetes, hypertension_history, cardiovascular_history

Clinical Features	bmi, waist_to_hip_ratio, systolic_bp, diastolic_bp, heart_rate, cholesterol_total, hdl_cholesterol, ldl_cholesterol, triglycerides, glucose_fasting, glucose_postprandial, insulin_level, hba1c
Additional Variables	diabetes_risk_score, diabetes_stage
Target Variable	diagnosed_diabetes

3.2.2 Data Preprocessing

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning modeling.

The preprocessing phase included:

- Dataset inspection and validation.
- Identification of numerical, categorical, binary, and target variables.
- Detection of missing values and duplicate records.
- Removal of potential data leakage variables, including `diagnosed_diabetes` and `diabetes_risk_score`.
- Encoding of the target variable, `diabetes_stage`, into numerical class labels.
- Splitting the dataset into training, validation, and test sets using stratified sampling.
- Encoding of categorical variables using one-hot encoding.
- Standardization of numerical features using `StandardScaler`.

The preprocessing transformations were fitted using the training data only and then applied to the validation and test sets to prevent data leakage.

These steps ensured that the dataset was properly prepared for subsequent model development.

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the characteristics of the dataset and identify patterns and relationships among the variables.

The analysis included:

- Descriptive statistical analysis of numerical features.
- Distribution analysis of categorical variables.
- Analysis of the target variable and class distribution.
- Distribution analysis of numerical features using histograms and density plots.
- Outlier detection using boxplots and the Interquartile Range (IQR) method.
- Correlation analysis among numerical predictors and with the target variable.
- Identification of highly correlated features to assess potential multicollinearity.
- Visualization using histograms, boxplots, heatmaps, and count plots.

The EDA results provided insights into the data distributions, class characteristics, outliers, and relationships among variables, and helped guide subsequent modeling decisions.

3.3.1 Data Quality Assessment

Data quality is the bedrock of reliable machine learning. In medical datasets specifically, poor-quality data can propagate into biased predictions with direct patient safety implications. The data quality assessment in this study examined four dimensions: missing values, duplicate records, categorical cardinality, and binary feature validity.

a. Missing Value Analysis

A comprehensive missing value audit was conducted by computing the count and percentage of null entries across all 31 columns. The results were unambiguous: no missing values were detected in any column of the dataset. This is a favorable characteristic that eliminates the need for imputation strategies such as mean/median substitution, K-Nearest Neighbor imputation, or multiple imputation by chained equations (MICE), all of which introduce additional assumptions and potential bias into the modeling process.

While this result is positive for analytical purposes, it is worth noting that in real-world clinical datasets, missing data rates of 10–30% are common, particularly for laboratory tests that may not have been ordered for all patients. The completeness of this dataset reflects its synthetic origins and should be acknowledged as a departure from the complexities typical of clinical data curation.

b. Duplicate Record Detection

A row-level deduplication check was performed to identify any patient records that appear more than once in the dataset. The analysis confirmed that zero duplicate rows exist. This is particularly important in supervised learning contexts: duplicate records in the training set can bias the model by artificially overweighting certain patient profiles, while duplicates that span the train/test boundary would constitute a form of data leakage and artificially inflate evaluation metrics.

3.3.2 Target Variable Distribution

Analysis of the target variable, *Diabetes Stage*, revealed a highly imbalanced multi-class distribution across five categories: No Diabetes, Pre-Diabetes, Type 1 Diabetes, Type 2 Diabetes, and Gestational Diabetes. Type 2 Diabetes was the predominant category, comprising 59,774 patients (59.8%) of the dataset, followed by Pre-Diabetes with 31,845 patients (31.8%). In contrast, No Diabetes accounted for 7,981 patients (8.0%), while Gestational Diabetes and Type 1 Diabetes were substantially underrepresented, with 278 (0.3%) and 122 patients (0.1%), respectively.

This distribution indicates a severe class imbalance, particularly for the Type 1 and Gestational Diabetes categories. Such imbalance can cause classification models trained on the original data distribution to favor the majority classes, potentially resulting in strong overall accuracy while providing inadequate predictive performance for minority classes. Therefore, model evaluation should not rely on overall accuracy alone, but should also consider class-sensitive metrics, including Macro-averaged Precision, Recall, and F1-score, which assign equal importance to each class. In addition, imbalance-handling techniques such as SMOTE, class weighting, or appropriate resampling strategies can be incorporated during model development to improve the representation and learning of minority classes.

The target variable, *Diabetes Stage*, was excluded from the predictor variables to prevent direct target leakage. Furthermore, variables that could introduce information leakage, including *diagnosed_diabetes* and *diabetes_risk_score*, were excluded from predictive modeling. These variables were retained only for descriptive analysis where appropriate.

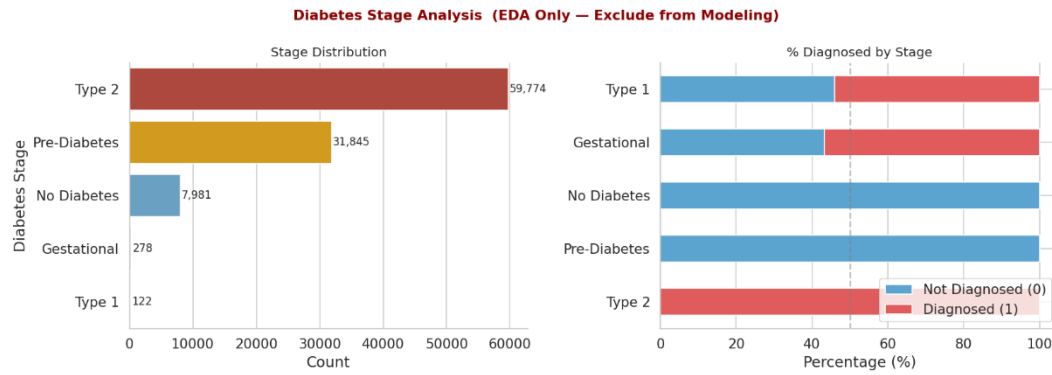


Fig. (3.1) Target variable distribution and diagnosis percentage across diabetes stages.

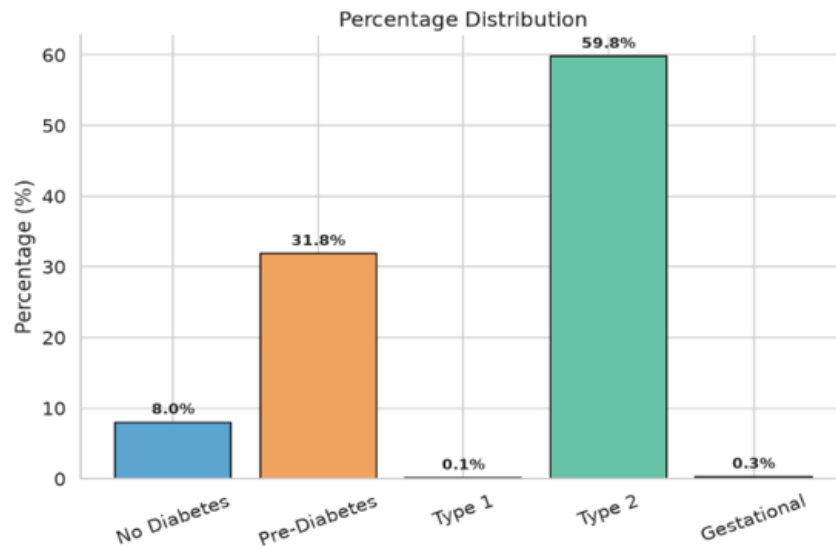


Fig. (3.2) Percentage distribution of diabetes categories

3.3.3 Demographic and Categorical Feature Analysis

Demographic and categorical feature analysis was conducted to examine the composition of the study population and assess the distribution of key sociodemographic and behavioral characteristics. The analysis included six categorical features: gender, ethnicity, education level, income level, employment status, and smoking status.

a. Distribution of Categorical Variables

The gender distribution was relatively balanced, with females representing approximately half of the study population and males accounting for a slightly smaller proportion, while a small proportion of

records belonged to other gender categories. The ethnic composition of the dataset was diverse, with White individuals representing the largest group, followed by Hispanic, Black, Asian, and other ethnic groups.

Educational attainment was primarily concentrated among individuals with high school and university-level education. Similarly, the analysis of employment status indicated that most participants were employed. Income level and smoking status were also examined to provide a broader understanding of the socioeconomic and behavioral characteristics of the study population.

Count plots were generated for all categorical features to assess the frequency and distribution of observations across their respective categories. The analysis showed generally reasonable representation across the categorical variables, with no severe underrepresentation observed in a single category. This distribution provides a useful basis for subsequent analysis and model development by allowing the relationships between sociodemographic and behavioral characteristics and diabetes stage to be further investigated.

Overall, the categorical feature analysis provided an overview of the demographic, socioeconomic, and behavioral characteristics of the dataset and supported the interpretation of patterns identified during the subsequent exploratory and predictive analyses.

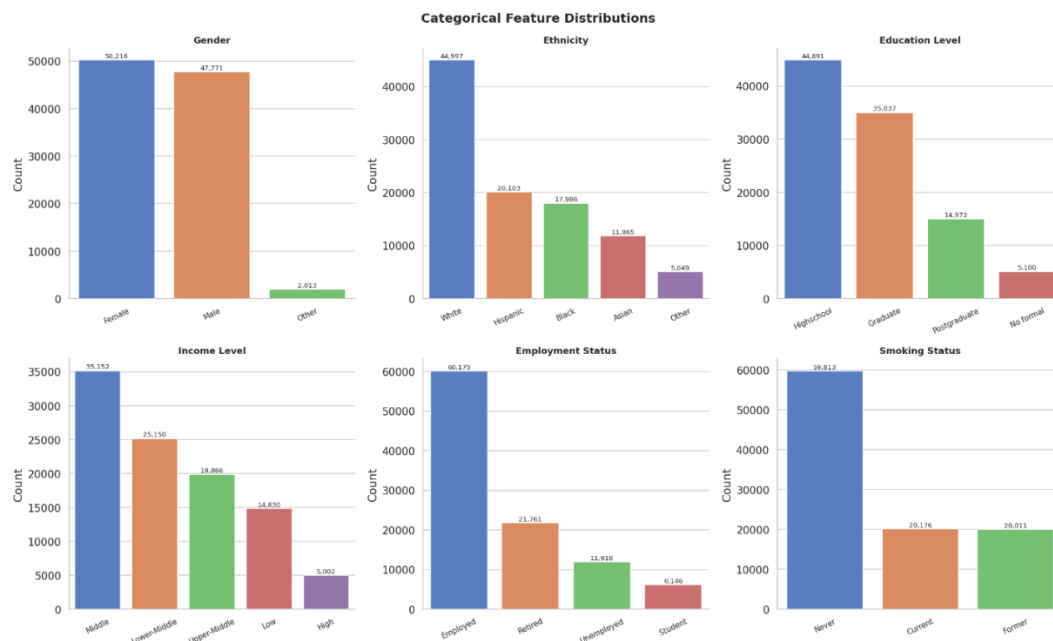


Fig. (3.3) Distributions of demographic and socioeconomic features

b. Binary Medical History Features

The three binary medical history features — `family_history_diabetes`, `hypertension_history`, and `cardiovascular_history` — received particular analytical attention due to their strong clinical relevance. These features represent established risk factors that have been extensively studied in the epidemiological literature.

Cross-tabulation and stratified distribution analysis across the five diabetes stages revealed that a positive history (Has History 1) in any of these three domains significantly shifts the patient profile toward more advanced diabetes stages, particularly Type 2 Diabetes (represented by the purple bars). This finding is clinically consistent with the pathophysiological mechanisms linking these conditions:

- Family History of Diabetes:** Genetic predisposition stands out as a dominant driver for disease progression. For individuals with no family history, Type 2 diabetes accounts for 54.7% of the subgroup, with Pre-Diabetes at 35.2% and No Diabetes at 9.8%. However, upon the presence of a family history, the proportion of Type 2 diabetes escalates sharply to 78.0%, while the normoglycemic (No Diabetes) fraction drops to a mere 1.6%. This is highly attributable to shared genetic susceptibility variants regulating insulin secretion and sensitivity (Morris et al., 2012).
- Hypertension History:** Hypertension and Type 2 diabetes co-occur at rates substantially exceeding chance. In patients with a positive history of hypertension, Type 2 diabetes prevalence increases from 59.0% to 62.2%, while Pre-Diabetes drops slightly to 30.6% and No Diabetes contracts to 7.0%. They share common etiological pathways including insulin resistance, endothelial dysfunction, and sympathetic nervous system overactivation, making this cluster a core component of metabolic syndrome (Grundy et al., 2005).
- Cardiovascular Disease History:** The bidirectional relationship between cardiovascular disease and diabetes is clearly reflected in the data. Patients with an existing history of cardiovascular disease show a marked expansion in Type 2 diabetes representation, rising from 59.3% to 64.9%, alongside a reduction in the Pre-Diabetes (28.6%) and No Diabetes (6.3%) cohorts. Hyperglycemia accelerates atherosclerosis, while cardiovascular inflammation contributes to peripheral insulin resistance, identifying this group as highly vulnerable (Emerging Risk Factors Collaboration, 2010).

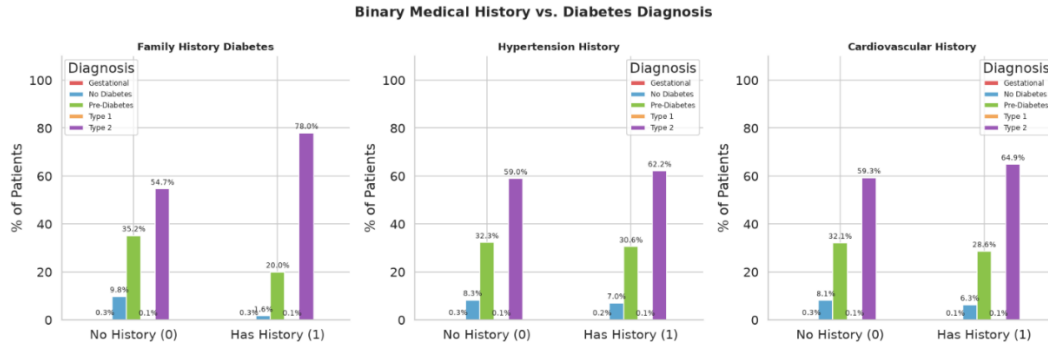


Fig. (3.4) Diabetes diagnosis distribution by medical history factors.

3.3.4 Numerical Feature Analysis

The numerical features represent an important source of clinical and behavioral information in the dataset. They include physiological, demographic, and lifestyle-related measurements that may provide useful information for distinguishing between the different diabetes-stage categories.

a. Distributional Analysis

Histograms were generated for the numerical features, with observations stratified by *Diabetes Stage*: Type 2 Diabetes, No Diabetes, Pre-Diabetes, Gestational Diabetes, and Type 1 Diabetes. This visualization enables comparison of the feature distributions across the different diabetes-stage categories and provides insight into the extent of distributional separation or overlap between groups. Several notable distributional patterns were observed:

- **Fasting Glucose and Postprandial Glucose:** These features showed the clearest distributional separation among the diabetes-stage categories. The No Diabetes group was generally concentrated at lower glucose levels, followed by Pre-Diabetes at intermediate levels, whereas Type 1 and Type 2 Diabetes showed distributions shifted toward higher glucose values. This pattern suggests that fasting and postprandial glucose provide strong descriptive separation between several diabetes-stage categories.
- **Age:** The age distributions varied across the diabetes-stage categories. Type 2 Diabetes showed a greater concentration among middle-aged and older individuals, whereas Gestational Diabetes was concentrated within younger age groups. This difference reflects the distinct age profiles observed across the categories in the dataset.

- **Body Mass Index (BMI):** The Type 2 Diabetes group showed a general tendency toward higher BMI values compared with the No Diabetes group. However, considerable overlap was observed among the five categories, indicating that BMI alone does not provide clear separation between diabetes stages.
- **Other Clinical and Lifestyle Features:** Variables such as systolic and diastolic blood pressure, cholesterol measures, physical activity, and diet-related scores showed substantial overlap across the diabetes-stage categories. Although these features may contribute useful information when considered jointly with other predictors, their individual distributions did not demonstrate strong separation between the classes.

Overall, the distributional analysis indicates that glucose-related measurements and age exhibit more pronounced differences across the diabetes-stage categories, while several other numerical features show considerable overlap. These findings provide useful descriptive insights for subsequent feature evaluation and predictive modeling, while recognizing that the discriminative contribution of each feature should be assessed further through statistical analysis and model-based evaluation.

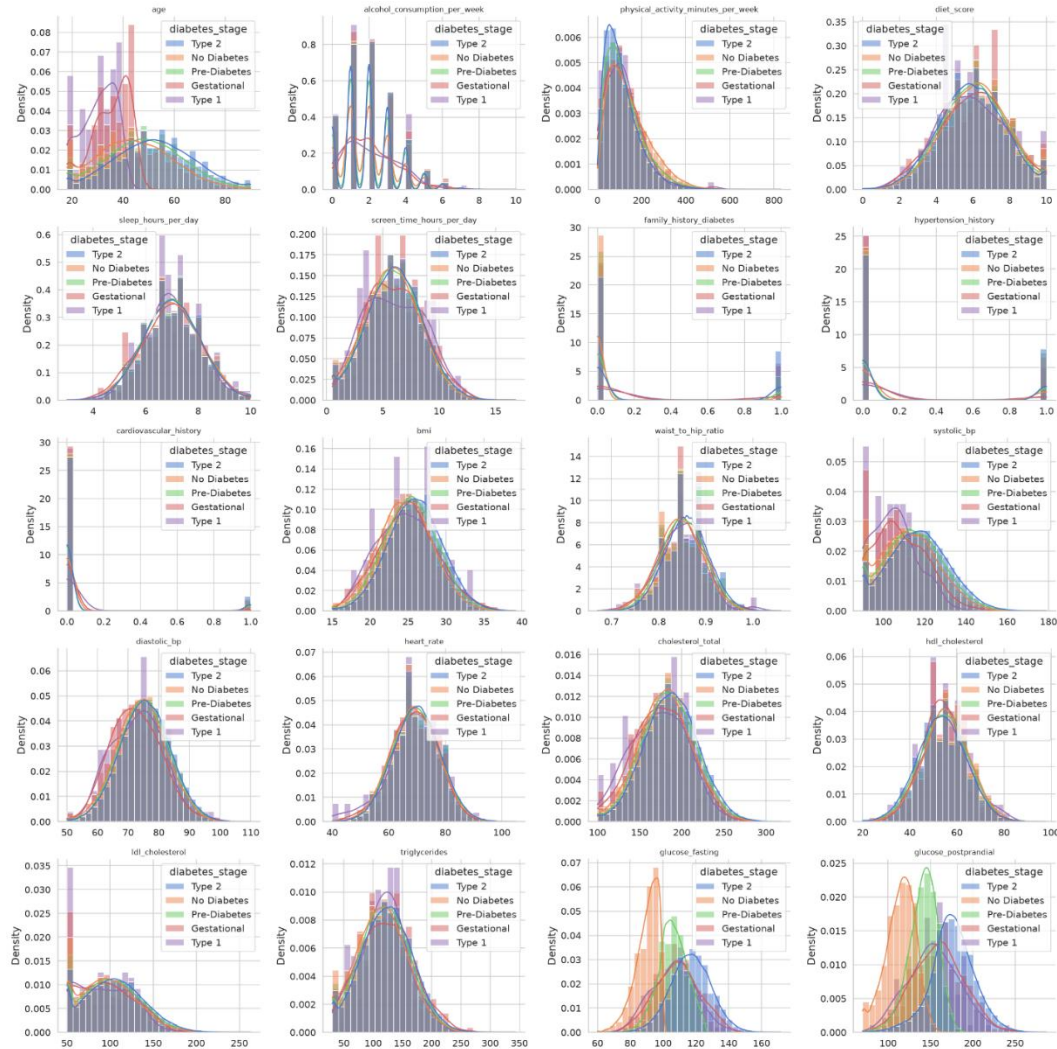


Fig. (3.5) Feature Distributions across Diabetes Stages

b. Discriminative Assessment via Stratified Boxplots

To further examine differences in the numerical features across the five diabetes-stage categories, stratified boxplots were constructed for each continuous variable. These visualizations provide a comparative view of the median, interquartile range (IQR), overall variability, and potential outliers within each category.

Several notable patterns were observed:

- **Glycemic Markers (HbA1c, Fasting Glucose, and Postprandial Glucose):** These features showed the clearest differences in their central distributions across the diabetes-stage categories. The No Diabetes group generally exhibited lower median values, while Pre-Diabetes showed

intermediate distributions. The Type 1 and Type 2 Diabetes groups tended to have higher median glucose-related measurements. This pattern indicates that glycemic markers provide strong descriptive separation among several of the target categories.

- **Age:** The boxplots revealed noticeable differences in age distributions across the diabetes-stage categories. Type 2 Diabetes showed a higher median age compared with the younger-oriented categories, while Gestational Diabetes exhibited a comparatively lower and narrower age distribution. Type 1 Diabetes also showed a younger age distribution than Type 2 Diabetes. These differences suggest that age may provide useful information for distinguishing between some of the categories.
- **Cardiovascular and Lipid Features:** Systolic and diastolic blood pressure, total cholesterol, and triglycerides generally showed higher median values and, in some cases, greater variability within the Type 2 Diabetes group. However, considerable overlap remained across the categories, indicating that these features provide less distinct separation when considered individually.
- **Outlier Distribution:** Potential statistical outliers were observed across several features, particularly Physical Activity, Insulin Level, and BMI. These observations were not automatically removed solely because they fell outside the boxplot whiskers. Instead, their values were considered in the context of the overall data distribution and feature characteristics. Where no clear evidence of data-entry errors or implausible measurements was identified, the observations were retained to avoid removing potentially informative variability from the dataset.

Overall, the stratified boxplots reinforce the findings from the distributional analysis. Glycemic markers and age demonstrated relatively clearer differences across the diabetes-stage categories, whereas features such as blood pressure, cholesterol, BMI, and physical activity showed greater overlap. These observations provide useful exploratory evidence for subsequent feature analysis and predictive modeling, but they do not by themselves establish the diagnostic or predictive importance of individual variables.

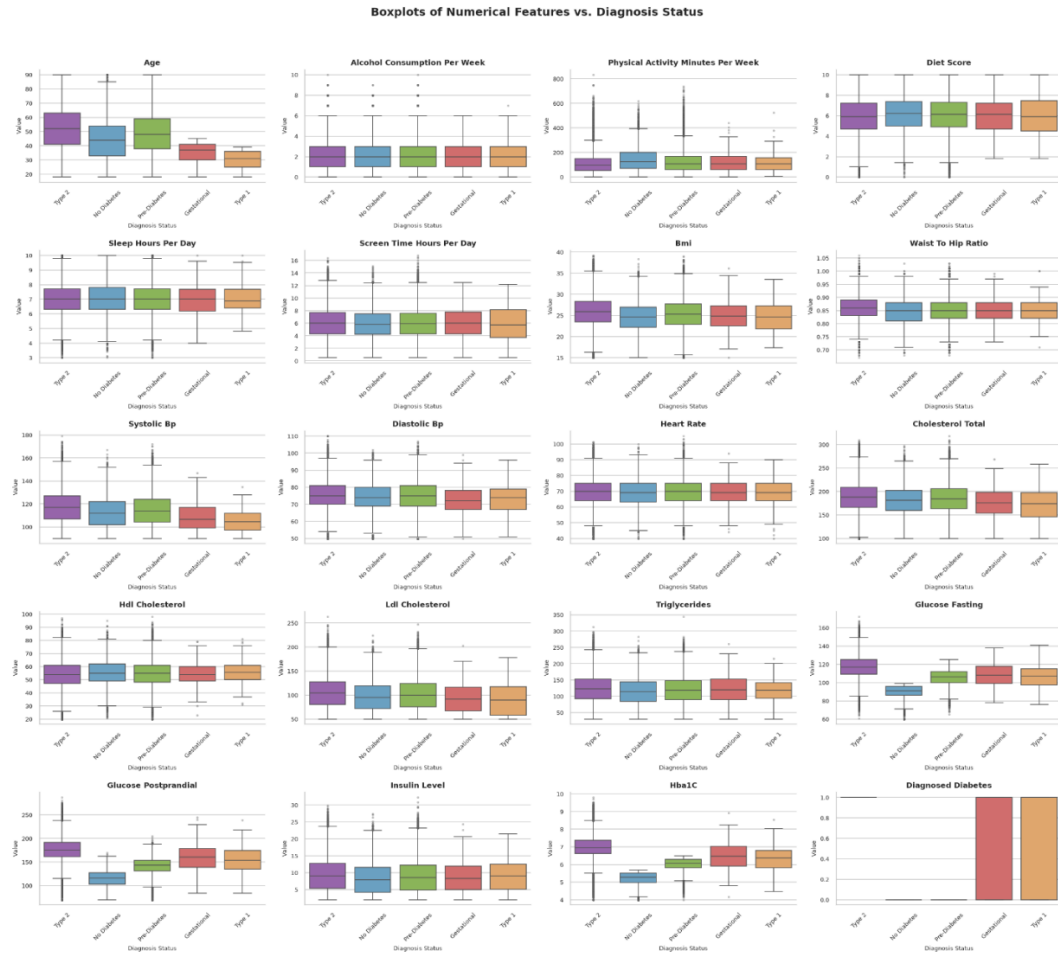


Fig. (3.6) Boxplots of Numerical Features Across Diabetes Stages

3.3.5 Correlation Analysis

Categorical features capture social determinants of health — a class of non-biological factors that have been increasingly recognized as powerful predictors of chronic disease risk. These include socioeconomic status, educational attainment, employment, and behavioral patterns such as smoking and alcohol consumption. Their analysis in this EDA serves two purposes: understanding the demographic composition of the study population and identifying subgroups with disproportionate diabetes prevalence.

a. Correlation Analysis

Correlation analysis was conducted to examine the linear relationships among numerical features and between selected numerical variables and target-related variables. This analysis provided insights into feature redundancy, potential multicollinearity, and the strength and direction of individual linear associations with the target.

Pearson correlation coefficients were used to quantify the strength and direction of linear relationships between numerical variables. Higher absolute correlation values indicate stronger linear associations, whereas values close to zero indicate weak linear relationships.

b. Feature-Target Correlation

Pearson correlation coefficients were calculated between the numerical features and the target-related variables *diagnosed_diabetes* and *Diagnosis_encoded*. The results indicate that glycemic markers exhibit the strongest individual linear associations with the target-related variables.

- **HbA1c** showed the strongest association, with a correlation of **0.68** with *diagnosed_diabetes* and **0.71** with *Diagnosis_encoded*.
- **Postprandial Glucose** followed with correlations of **0.63** and **0.66**, respectively.
- **Fasting Glucose** showed correlations of **0.51** with *diagnosed_diabetes* and **0.55** with *Diagnosis_encoded*.

These results are consistent with the distributional analysis, where glycemic measurements also demonstrated clearer differences across the diabetes-stage categories.

The variable *diabetes_risk_score* showed a correlation of **0.28** with *diagnosed_diabetes* and was also correlated with several risk-related features, including Age (**0.50**) and Fasting Glucose (**0.47**). More importantly, *Diagnosis_encoded* exhibited an extremely high correlation of **0.98** with *diagnosed_diabetes*. Since these variables directly or indirectly represent diagnosis-related information, they were excluded from the predictive feature set to prevent data leakage.

Among the variables considered safe for predictive modeling, Age showed a relatively weak positive correlation with the target-related variables (**0.14** with *diagnosed_diabetes* and **0.15** with *Diagnosis_encoded*). *Physical_activity_minutes_per_week* showed a weak negative correlation of **-0.10** and **-0.11**, respectively. These relatively small coefficients indicate that the individual linear relationships of these variables with the target are limited, although they may still provide useful information when combined with other predictors in multivariate models.

c. Correlation Heatmap and Multicollinearity

A full Pearson correlation matrix was visualized using a lower-triangle heatmap to examine relationships among numerical features and identify potential multicollinearity.

Several notable relationships were observed:

- **Lipid Features:** *cholesterol_total* and *ldl_cholesterol* showed a strong positive correlation of **0.91**, indicating substantial redundancy between the two variables. Including highly correlated predictors may introduce multicollinearity, particularly in models whose coefficients are sensitive to correlated inputs.
- **Glycemic Features:** Strong correlations were observed among the glycemic markers. *HbA1c* correlated with *glucose_postprandial* at **0.93** and with *glucose_fasting* at **0.70**, while *glucose_fasting* and *glucose_postprandial* showed a correlation of **0.59**. These relationships indicate that the glycemic variables contain substantial overlapping linear information.
- **Adiposity and Physiological Features:** *BMI* and *waist_to_hip_ratio* showed a strong positive correlation of **0.77**, indicating considerable overlap between these measures of body composition. In addition, *systolic_bp* showed a moderate positive correlation of **0.55** with Age.
- **Risk-Related Features:** *family_history_diabetes* and *diabetes_risk_score* also showed a strong positive association, reflecting the contribution of family history to the composite risk measure.

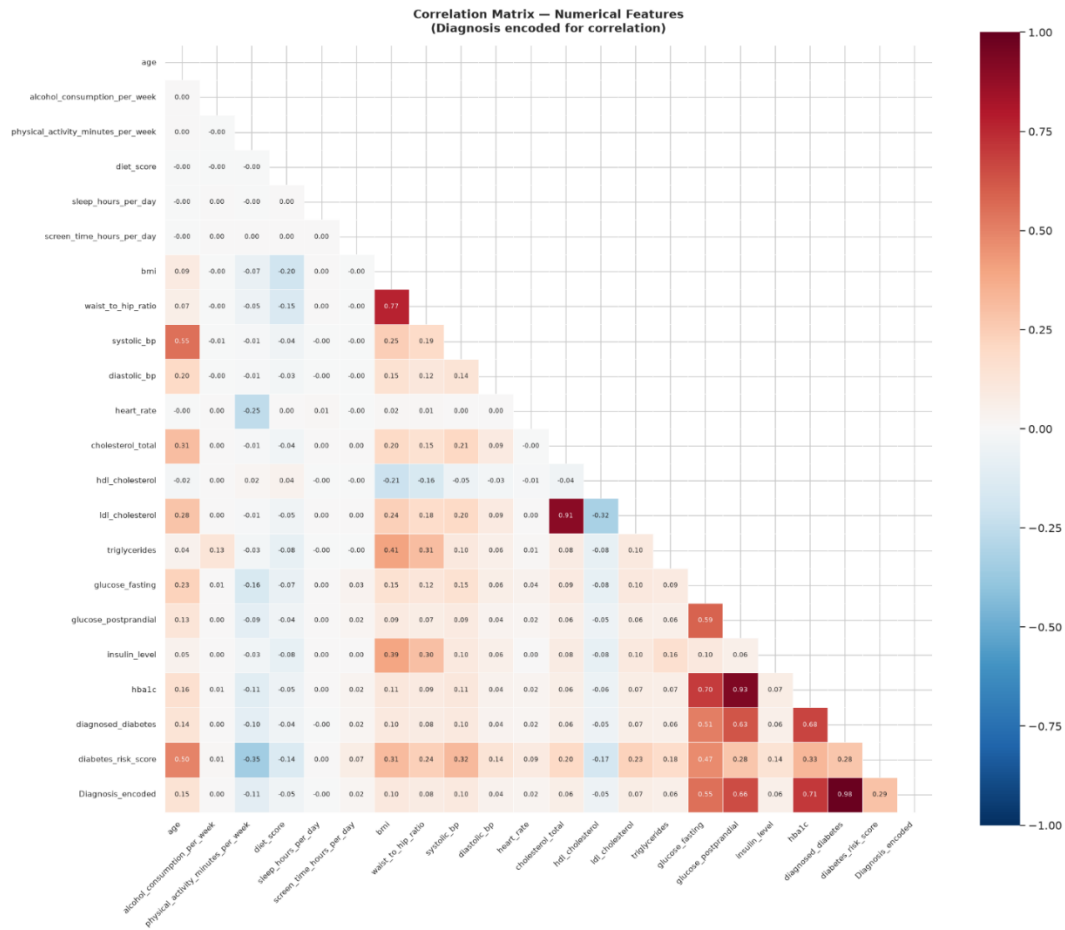


Fig. (3.7) Correlation matrix of numerical features

Feature pairs with an absolute Pearson correlation coefficient **greater than 0.70** were systematically identified to assess potential redundancy among the numerical predictors. The identified high-correlation pairs included *glucose_postprandial* and *HbA1c*, *cholesterol_total* and *ldl_cholesterol*, *BMI* and *waist_to_hip_ratio*, and *family_history_diabetes* and *diabetes_risk_score*. These relationships indicate that several predictors capture overlapping information.

The presence of highly correlated predictors is particularly relevant for linear models, where multicollinearity can increase the variance of coefficient estimates and make individual coefficients less stable and more difficult to interpret. In contrast, tree-based models such as Random Forest and XGBoost are generally less sensitive to multicollinearity, although redundant features may still affect feature-importance interpretation.

Therefore, highly correlated features were not automatically removed based solely on Pearson correlation. Instead, their redundancy was considered during subsequent feature selection, with **Variance**

Inflation Factor (VIF) analysis used as an additional diagnostic for assessing multicollinearity before applying linear models.

Overall, the correlation analysis showed that **glycemic, lipid, and adiposity-related variables contained several strong inter-feature relationships**, while glycemic markers also demonstrated the strongest individual linear associations with the target. These findings were used as exploratory evidence for subsequent feature selection and multicollinearity assessment rather than as the sole basis for removing predictors.

3.3.6 Findings and Implications for Modeling

The exploratory data analysis (EDA) phase revealed several key findings that informed the subsequent machine learning strategy.

- **Primary Predictor Hierarchy:** Glycemic biomarkers, particularly **HbA1c, Postprandial Glucose, and Fasting Glucose**, demonstrated the strongest individual associations with the target-related variables and showed clear differences across diabetes-stage categories. These findings indicate that glycemic measurements are likely to play a major role in distinguishing diabetes-related classes within the dataset.
- **Socio-Demographic and Risk-Related Patterns:** **Family history of diabetes** showed a strong association with the composite *diabetes_risk_score*, while **Age** demonstrated distinct distributional patterns across some diabetes-stage categories. These findings suggest that demographic and risk-related variables may provide complementary information beyond glycemic measurements.
- **Multicollinearity Considerations:** Several highly correlated predictor pairs were identified, particularly among **glycemic, lipid, and adiposity-related features**. The strongest relationships included *cholesterol_total* with *ldl_cholesterol* ($r = 0.91$), *HbA1c* with *glucose_postprandial* ($r = 0.93$), and *BMI* with *waist_to_hip_ratio* ($r = 0.77$). These relationships indicate potential feature redundancy and were therefore considered during subsequent feature-selection and multicollinearity assessment, particularly for linear models.
- **Leakage Prevention:** The analysis identified *Diagnosis_encoded* as highly correlated with *diagnosed_diabetes* ($r = 0.98$) and identified *diabetes_risk_score* as a diagnosis-related variable with potential information leakage. Consequently, these variables were excluded from the predictive feature set before model training to prevent artificially inflated performance and ensure a more reliable evaluation of model generalization.

Overall, the EDA established that **glycemic measurements represent the strongest individual predictors**, while demographic, anthropometric, physiological, and lifestyle-related variables may provide additional predictive information. At the same time, the presence of correlated predictors and diagnosis-related variables highlighted the need for **careful feature selection, leakage prevention, and model-specific handling of multicollinearity** in the subsequent modeling stages.

3.3.7 Advanced Feature Selection and Non-Linear Relationships

Although Pearson correlation is useful for assessing linear associations between numerical features and the target, it may not capture non-linear statistical dependencies. Therefore, **Mutual Information (MI)** was used as a complementary criterion to assess the statistical dependency between each feature and the target.

The MI analysis revealed a clear hierarchy of feature–target dependencies. **HbA1c** showed the highest MI score (**0.4417**), followed by **Glucose Postprandial (0.2884)** and **Glucose Fasting (0.1688)**. Among non-glycemic variables, **Family History of Diabetes (0.0239)** and **Age (0.0111)** showed the highest MI scores. Several demographic, behavioral, anthropometric, and lifestyle variables exhibited substantially lower MI values, including **Smoking Status – Never (0.0084)** and **BMI (0.0062)**.

These results confirmed the dominant individual information content of glycemic biomarkers and helped identify features with relatively low univariate statistical dependency on the target. However, low MI values were not interpreted as evidence that a feature was completely irrelevant, since weak individual associations may still contribute through multivariate interactions.

The MI results were subsequently used to guide the feature selection and engineering stage described in **Section 3.4**.

3.4 Feature Engineering Pipeline

Following the initial baseline modeling experiments, a feature engineering and selection stage was performed to refine the feature representation. The initial One-Hot Encoded representation contained **40 features**.

Based on the MI analysis, **nine low-information features** were removed: *diastolic_bp*, *screen_time_hours_per_day*, *diet_score*, *alcohol_consumption_per_week*, *education_level_No formal*, *education_level_Postgraduate*, *income_level_Upper-Middle*, *employment_status_Unemployed*, and *employment_status_Student*.

In addition, a clinically meaningful interaction feature, the **Postprandial-to-Fasting Glucose Ratio** (*postprandial_to_fasting_ratio*), was created to capture the relationship between fasting and post-meal glucose levels.

The removal of the nine features reduced the feature space from **40 to 31 features**, while the addition of the engineered ratio resulted in a final feature space of **32 features**.

The resulting feature representation was subsequently used for the second-stage experiments involving **SVM, MLP, and XGBoost**.

3.5 Preparing Machine Learning Models

The modeling process was conducted in two stages. First, five machine learning algorithms—**Support Vector Machine (SVM), Random Forest (RF), XGBoost, LightGBM, and Multi-Layer Perceptron (MLP)**—were trained using the initial feature representation to establish a baseline comparison.

Following the baseline experiments and the feature engineering process described in Section 3.4, **SVM, MLP, and XGBoost** were selected for the subsequent experimental stages based on their comparative performance and complementary learning characteristics.

To address class imbalance, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied exclusively to the training data, while the validation and test sets remained untouched to ensure unbiased performance evaluation.

Hyperparameter optimization was performed using **GridSearchCV with stratified cross-validation** to systematically evaluate predefined parameter combinations for the selected models. SMOTE was applied only to the corresponding training data to prevent information leakage during model optimization.

All subsequent model evaluations were conducted using the same validation and test partitions and consistent performance metrics.

3.6 Machine Learning Models



Fig. (3.8) Machine Learning Algorithms Used for Classification

The modeling pipeline incorporates a diverse set of classifiers spanning kernel-based, tree-based, gradient-boosting, and neural-network architectures. This diversity enables comparative evaluation across different learning strategies and inductive biases for the structured tabular diabetes dataset. The following subsections describe each selected algorithm and the rationale for its inclusion.

3.6.1 Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm that identifies a decision boundary that maximizes the margin between classes. In this study, SVM was configured with an RBF (Radial Basis Function) kernel to capture non-linear decision boundaries and was optimized using cross-validation. SVM was included because of its ability to model complex class boundaries and its effectiveness in high-dimensional feature spaces.

3.6.2 Random Forest (RF)

Random Forest is an ensemble learning algorithm that constructs multiple decision trees using randomized subsets of the training data and aggregates their predictions through majority voting. The use of **bootstrap aggregation (bagging)** helps reduce variance and limit overfitting. Random Forest was selected because of its strong performance on structured tabular data, robustness to noisy features, and ability to provide feature-importance measures for subsequent interpretation.

3.6.3 Extreme Gradient Boosting (XGBoost)

XGBoost is a regularized implementation of gradient boosting that builds decision trees sequentially, with each subsequent tree focusing on correcting the errors of the preceding ensemble. The algorithm incorporates L1 and L2 regularization to control overfitting while effectively modeling non-linear relationships and feature interactions. XGBoost was selected for its strong predictive performance and computational efficiency on tabular datasets.

3.6.4 Light Gradient Boosting Machine (LightGBM)

LightGBM is a gradient-boosting framework developed by Microsoft that uses a **leaf-wise tree growth strategy**, allowing computational resources to focus on splits that provide greater improvement in the objective function. This approach can provide efficient training and reduced memory usage, particularly for large datasets. LightGBM was included to evaluate an alternative gradient-boosting strategy for modeling complex relationships within the clinical and demographic features.

3.6.5 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) is a feedforward neural network composed of fully connected layers that learn non-linear relationships through weighted transformations and activation functions. In this study, the network was trained using the **Adam optimizer** and **Sparse Categorical Crossentropy** loss, which is appropriate for multi-class classification with integer-encoded labels. MLP was included to compare neural-network-based learning with the tree-based and kernel-based approaches used in this study.

3.6.6 Model Explainability and Experiment Tracking

Beyond predictive performance, **model explainability** was incorporated to examine the contribution of individual features to model predictions. **SHAP (SHapley Additive exPlanations)** was employed to provide interpretable explanations of model behavior at both the global and individual prediction levels. This enabled the identification of the features that contributed most strongly to the model's predictions and allowed individual patient predictions to be examined in a more transparent manner.

All experimental runs were tracked using **MLflow**, including model configurations, hyperparameters, evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC, as well as relevant model artifacts. This systematic experiment-tracking process supports consistent model comparison, reproducibility, and reliable documentation of the experimental results.

Table (3.2) General Common Hyperparameters of ML models

Hyperparameter	Models Using It	Purpose
Random State	SVM, XGBoost, MLP, LightGBM, Random Forest	Ensures reproducibility by controlling the randomness during model training and data sampling.
Learning Rate	XGBoost, LightGBM, MLP	Controls the step size of model updates during training and affects the convergence speed.
Number of Estimators (n_estimators)	XGBoost, LightGBM, Random Forest	Determines the number of trees used in ensemble learning methods.
Maximum Depth (max_depth)	XGBoost, LightGBM	Controls the complexity of decision trees and helps prevent overfitting.
Regularization Parameters	SVM (C), MLP (Alpha)	Reduce overfitting and improve model generalization performance.
Maximum Iterations (max_iter)	MLP	Defines the maximum number of training iterations required for convergence.

Table (3.3) Model Hyperparameters

Model	Parameters
SVM	C, Kernel, Gamma, Probability, Random State
XGBoost	n_estimators, max_depth, learning_rate, subsample, colsample_bytree, eval_metric, random_state
MLP	hidden_layer_sizes, activation, solver, alpha, learning_rate_init, max_iter, random_state
LightGBM	n_estimators, learning_rate, num_leaves, max_depth, random_state
Random Forest	n_estimators, random_state, n_jobs

3.7 Adapted RAG Pipeline

Retrieval-Augmented Question Answering over a Clinical Nutrition Text (Nutritional Management of Diabetes Mellitus)

1. Overview

This project implements a fully offline, locally-run Retrieval-Augmented Generation (RAG) system that answers clinical questions grounded in a single source document: *Nutritional Management of Diabetes Mellitus* (Frost, 2003, Wiley), a 271-page reference. The pipeline was built and executed as a 22-stage Jupyter notebook covering ingestion, hybrid retrieval, reranking, local LLM generation, and automated evaluation — with no external API calls, so all inference runs on local hardware via Ollama.

2. Pipeline Architecture

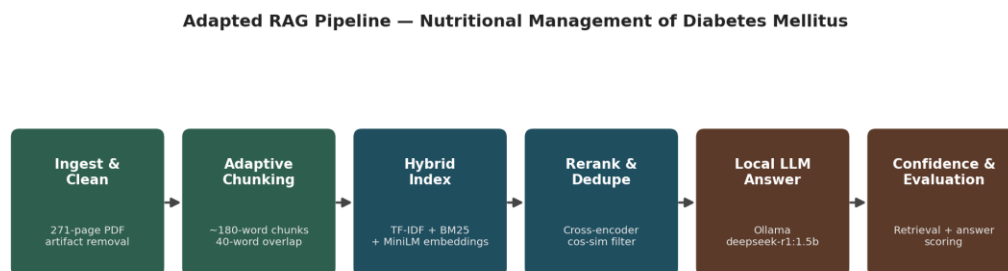


Fig. (3.9) End-to-End Architecture of the Adapted RAG Pipeline

3. Stage-by-Stage Summary

- Ingestion & Cleaning (Stages 1–5): pypdf extracts text page-by-page; a cleaning pass removes hyphenation artifacts, Wiley publisher boilerplate, headers/footers, URLs and page numbers, then filters out low-content front-matter pages (<50 words).
- Adaptive Chunking (Stages 6–7): paragraph-aware splitting into ~180-word chunks with 40-word overlap, followed by a quality-validation pass to drop degenerate chunks.
- Hybrid Indexing (Stages 9–13): three parallel retrieval signals — TF-IDF cosine similarity, a custom BM25 implementation ($k_1=1.5$, $b=0.75$), and dense semantic embeddings (Sentence-

Transformers “all-MiniLM-L6-v2”) — combined with a weighted fusion score (TF-IDF 0.25 / BM25 0.25 / semantic 0.50) plus lexical query expansion.

- **Reranking & Deduplication (Stages 14–16):** a cross-encoder (“cross-encoder/ms-marco-MiniLM-L12-v2”) re-scores the top candidates for query–passage relevance; near-duplicate chunks (cosine similarity ≥ 0.90) are then removed from the final context.
- **Context, Prompting & Generation (Stages 15, 17–18):** the surviving top chunks are assembled into a source-cited context block and passed, inside a constructed prompt, to a local Ollama model (deepseek-r1:1.5b, temperature 0.1) for answer generation.
- **Grounding & Confidence (Stages 19–20):** answers below a rerank-score threshold (1.0) are withheld with a low-confidence notice instead of being generated, and every answer is tagged with a High/Medium/Low confidence label derived from the reranker scores.
- **Evaluation (Stages 21–22):** automated retrieval statistics (score spread, unique source pages) and an embedding-based answer-context cosine similarity check, so answer quality is measured rather than assumed.

4. Sample Run — Key Metrics

Table (3.4) Key Evaluation Metrics for Sample

Metric	Value	Interpretation
Pages processed	271 (0 needing OCR)	Clean, fully machine-readable source text
Best rerank score	2.788	High-confidence retrieval band (≥ 2.5)
Average rerank score	1.626	Consistent relevance across returned chunks
Answer-context similarity	0.771	Good semantic alignment between answer and evidence

Tech stack: Python, pypdf, scikit-learn (TF-IDF), a custom BM25 index, Sentence-Transformers (bi-encoder + cross-encoder), pandas, and Ollama running deepseek-r1:1.5b locally. [26]

CHAPTER 4

Results and Discussion

4.1 Introduction

This chapter presents the practical implementation and experimental evaluation of the proposed explainable machine learning framework for diabetes prediction. Building upon the methodology described in Chapter 3, the complete prediction pipeline was implemented using the Diabetes Healthcare Dataset through sequential stages of data preprocessing, baseline model development, feature engineering, and class-imbalance handling.

Multiple machine learning models, including Support Vector Machine (SVM), Random Forest (RF), XGBoost, LightGBM, and Multi-Layer Perceptron (MLP), were trained and systematically evaluated under consistent experimental conditions. In addition, a separate Neural Network (NN) architecture was developed using TensorFlow/Keras. Model performance was assessed using Accuracy, Precision, Recall, F1-score, and ROC-AUC to provide a comprehensive comparison of predictive performance.

Principal Component Analysis (PCA) was used to examine class separability and visualize the feature space, while MLflow was employed for experiment tracking and reproducibility. SHAP was incorporated to provide interpretable explanations of model predictions and quantify the contribution of individual features.

Finally, the selected best-performing model was integrated into an interactive Streamlit-based dashboard for patient-level diabetes prediction and explanation. The chapter concludes with a comparative analysis of the implemented models and discusses the selection of the most suitable model for deployment within the proposed system.

4.2 System Implementation

4.2.1 Development Environment

The development environment used in this study was designed to support the implementation, training, evaluation, interpretation, and deployment of the proposed diabetes prediction framework. The complete workflow was developed using Python due to its extensive ecosystem of libraries for data preprocessing, machine learning, visualization, explainability, and web application development.

All experiments were conducted within the Jupyter Notebook environment, which facilitated interactive data analysis, model development, and result visualization. Data preprocessing, feature engineering, and data manipulation were performed using Pandas and NumPy, while Matplotlib and Seaborn were used for data visualization. Principal Component Analysis (PCA) was implemented using Scikit-learn to examine the distribution of the feature space and assess class separability.

The initial machine learning models, including Support Vector Machine (SVM), Random Forest (RF), and Multi-Layer Perceptron (MLP), were implemented using Scikit-learn, while XGBoost and LightGBM were implemented using their respective frameworks. Following the initial comparative modeling stage, SVM, MLP, and XGBoost were selected for the subsequent experiments after feature engineering. In addition, a separate Neural Network (NN) architecture was developed using TensorFlow/Keras.

SMOTE was applied using the imbalanced-learn library to address class imbalance during model training. MLflow was used to track experimental runs, model parameters, evaluation metrics, and model artifacts, supporting reproducibility and systematic comparison. SHAP was used for model explainability to analyze feature contributions to individual predictions.

Finally, the selected best-performing model was integrated into an interactive Streamlit-based dashboard, allowing users to enter patient information and obtain diabetes predictions accompanied by interpretable feature-level explanations.

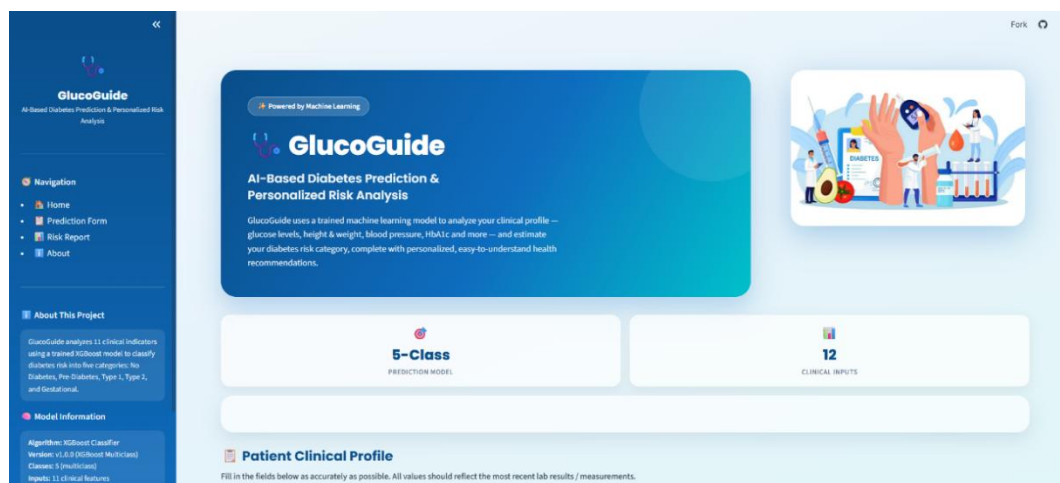


Fig. (4.1) Interactive Web-Based Diabetes Risk Assessment Dashboard (GlucoGuide).

4.3 Model Training

4.3.1 Data Splitting Strategy

The data splitting strategy adopted in this study was designed to support reliable model training, validation, and unbiased performance evaluation. Following the initial data preprocessing stage, the dataset was partitioned into three independent subsets using stratified sampling to preserve the class distribution across the different partitions.

The dataset was divided into training, validation, and evaluation partitions using a ratio of 80:15:5, respectively. Specifically, 80,000 samples (80%) were allocated to the training set, 15,000 samples (15%) were assigned to the validation set, and 5,000 samples (5%) were reserved for the evaluation set.

The training partition was used for model fitting, while the validation partition was used to monitor model performance and support model selection and hyperparameter optimization. The evaluation partition was kept unseen during the training and model-selection stages and was reserved for the final assessment of model generalization.

This splitting strategy was initially applied to support the baseline comparison of five machine learning algorithms: Support Vector Machine (SVM), Random Forest (RF), XGBoost, LightGBM, and Multi-Layer Perceptron (MLP). Following the baseline modeling stage, feature engineering and selection were performed, resulting in a refined feature representation of 32 features. SVM, MLP, and XGBoost were subsequently selected for the second-stage experiments.

4.3.2 Handling Class Imbalance (SMOTE)

During exploratory data analysis, the target variable was found to exhibit substantial class imbalance, with some diabetes stages being considerably underrepresented compared with the majority classes. Such imbalance may cause machine learning models to favor majority classes and reduce their ability to correctly identify minority classes. To investigate the effect of class-imbalance handling, several approaches were evaluated, including the Original (Imbalanced) dataset, Random Oversampling (ROS), SMOTE Optimization, and SMOTEENN Cleaned Optimization.

Each approach was applied to the training dataset, and the resulting models were evaluated under the same experimental conditions. As presented in Table X, the Original (Imbalanced) approach achieved the best overall performance across the evaluated metrics, with an Accuracy of 0.9169, Precision of 0.9263, Recall of 0.9169, F1-score of 0.9160, and the highest ROC-AUC of 0.9510.

Although Random Oversampling and SMOTE Optimization produced competitive results, neither method consistently outperformed the Original dataset across the evaluated metrics. SMOTEENN Cleaned Optimization resulted in lower performance, with an Accuracy of 0.9049 and ROC-AUC of 0.9388. The Original dataset therefore provided the strongest overall performance without requiring artificial resampling.

Based on these experimental findings, the Original (Imbalanced) dataset was selected as the preferred class-imbalance strategy for the subsequent modeling experiments..

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** Executing Imbalance Resampling Experiments...

```

	Resampling Technique	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Original (Imbalanced)	0.9169	0.9263	0.9169	0.9160	0.9510
3	Random Oversampling	0.9155	0.9265	0.9155	0.9154	0.9503
1	SMOTE Optimization	0.9163	0.9259	0.9163	0.9155	0.9493
2	SMOTEENN Cleaned Optimization	0.9049	0.9201	0.9049	0.9070	0.9388

Fig. (4.2) Performance Evaluation of Imbalance Resampling Techniques

4.4 Model Evaluation

4.4.1 Support Vector Machine (SVM)

The Support Vector Machine (SVM) classifier was implemented as one of the machine learning models evaluated in this study to assess its capability for diabetes-stage prediction using the processed healthcare dataset. The model was trained and evaluated under the same experimental conditions as the other selected models using the validation dataset and multiple classification metrics.

As shown in the validation results, the SVM model achieved an **Accuracy of 0.8693**, with a **Precision of 0.8740**, **Recall of 0.8693**, and **F1-score of 0.8695**. The close values of precision, recall, and F1-score indicate a relatively consistent overall classification performance. The model also achieved a **ROC-AUC of 0.9388**, indicating strong discriminative capability across the diabetes-stage classes.

Overall, the SVM demonstrated reliable predictive performance and provided a strong basis for comparison with the other machine learning models evaluated in this study.

```

Run Logged Successfully in MLflow

Validation Metrics
Accuracy : 0.8693
Precision: 0.8740
Recall   : 0.8693
F1 Score : 0.8695
ROC-AUC  : 0.9388

```

Fig. (4.3) MLflow Experiment Logging and Model Validation Metrics

4.4.2 Random Forest (RF)

The Random Forest (RF) classifier was implemented as an ensemble learning algorithm to evaluate its effectiveness in predicting diabetes stages using the processed healthcare dataset. Random Forest combines multiple decision trees through bootstrap aggregation, allowing the model to capture complex relationships within the feature space while reducing the risk of overfitting.

The validation results indicate that the Random Forest model achieved an **Accuracy of 0.9163**, with a **Precision of 0.9259**, **Recall of 0.9163**, and **F1-score of 0.9152**. The relatively close values of these metrics indicate consistent overall classification performance. The model also achieved a **ROC-AUC of 0.9486**, indicating strong discriminative capability across the diabetes-stage classes.

Overall, the Random Forest classifier demonstrated strong validation performance and provided a competitive result among the evaluated machine learning models. Its ensemble-based structure enables it to capture nonlinear relationships and interactions among the input features, making it a suitable model for diabetes-stage prediction.

```

Run Logged Successfully
=====
Run ID : 9a7b5636a013429c96e7770a221b4af9

Accuracy : 0.9163
Precision: 0.9259
Recall   : 0.9163
F1 Score : 0.9152
ROC-AUC  : 0.9486

```

Fig. (4.4) Random Forest Validation Performance

4.4.3 XGBoost

The Extreme Gradient Boosting (XGBoost) algorithm was implemented as one of the selected machine learning models due to its ability to capture complex nonlinear relationships and interactions within structured healthcare data. Following the initial baseline modeling stage, XGBoost was retained for the subsequent experimental stages together with SVM and MLP after the feature engineering and selection process.

The validation results demonstrated strong predictive performance. The XGBoost model achieved an **Accuracy of 0.9163**, with a **Precision of 0.9258**, **Recall of 0.9163**, and **F1-score of 0.9154**. The relatively close values of these metrics indicate consistent overall classification performance. The model also achieved a **ROC-AUC of 0.9512**, indicating strong discriminative capability across the diabetes-stage classes.

The strong performance of XGBoost can be attributed to its gradient boosting mechanism, which sequentially builds decision trees to reduce prediction errors and effectively model nonlinear relationships and feature interactions. The model therefore demonstrated competitive validation performance and was considered one of the strongest candidates for the subsequent stages of the proposed diabetes prediction framework.

```
-----  
Run Logged Successfully  
=====  
Run ID : 014f06407d6343b99fcdcef7d581d254  
  
Accuracy : 0.9163  
Precision: 0.9258  
Recall   : 0.9163  
F1 Score : 0.9154  
ROC-AUC  : 0.9512
```

Fig. (4.5) XGBoost Validation Performance

4.4.4 LightGBM

The Light Gradient Boosting Machine (LightGBM) was implemented to evaluate the effectiveness of gradient boosting techniques for diabetes-stage prediction. LightGBM is designed to efficiently process structured datasets while maintaining strong predictive performance and computational efficiency. The model was evaluated under the same experimental conditions as the other machine learning models.

The validation results demonstrate that the LightGBM model achieved an **Accuracy of 0.9170**, with a **Precision of 0.9262**, **Recall of 0.9170**, and **F1-score of 0.9160**. The relatively close values of these metrics indicate consistent overall classification performance. The model also achieved a **ROC-AUC of 0.9514**, indicating strong discriminative capability across the diabetes-stage classes.

Overall, LightGBM demonstrated strong validation performance and achieved competitive results compared with the other evaluated models. Its gradient boosting framework provides an effective approach for modeling complex relationships within structured healthcare data, making it a suitable candidate for diabetes-stage prediction.

```
=====
Run Logged Successfully
=====
Run ID : d28dfca9472a4ac2b7fb4341427a29c3

Validation Metrics
Accuracy : 0.9170
Precision: 0.9262
Recall   : 0.9170
F1 Score : 0.9160
ROC-AUC  : 0.9514
```

Fig. (4.6) LightGBM Validation Performance

4.4.5 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) classifier was implemented to investigate the capability of neural network-based learning in predicting diabetes stages using the proposed healthcare dataset. MLP was evaluated as one of the machine learning models under the same experimental conditions as the other classifiers.

The validation results indicate that the MLP model achieved an **Accuracy of 0.9047**, with a **Precision of 0.9123**, **Recall of 0.9047**, and **F1-score of 0.9042**. The relatively close values of these metrics indicate consistent overall classification performance. The model also achieved a **ROC-AUC of 0.9442**, demonstrating strong discriminative capability across the diabetes-stage classes.

The MLP classifier was able to model nonlinear relationships within the input feature space, resulting in competitive validation performance. Compared with the other evaluated machine learning models, MLP provided a strong neural-network-based benchmark and was subsequently retained for the second-stage experimental analysis alongside **SVM and XGBoost**.

```
Run Logged Successfully
=====
Run ID : 23f56aaf3e6047f8a1287e8bab3b2d1b

Accuracy : 0.9047
Precision: 0.9123
Recall    : 0.9047
F1 Score  : 0.9042
ROC-AUC   : 0.9442
```

Fig. (4.7) MLP Validation Performance

4.5 Comparative Analysis of Machine Learning Models

4.5.1 Evaluation Metrics

Comparison of Machine Learning Models

The performance of the five **baseline machine learning models** was evaluated using five widely accepted classification metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. These models were trained using the initial preprocessed feature representation **before the subsequent feature engineering stage**. The results provide a baseline for comparing the predictive capabilities of the different algorithms prior to feature refinement.

The Support Vector Machine (SVM) achieved the lowest overall performance among the evaluated models, with an Accuracy of 86.93%, a Precision of 87.40%, a Recall of 86.93%, an F1-Score of 86.95%, and a ROC-AUC of 93.88%. The Multi-Layer Perceptron (MLP) demonstrated improved performance compared to SVM, achieving an Accuracy of 90.47%, a Precision of 91.23%, a Recall of 90.47%, an F1-Score of 90.42%, and a ROC-AUC of 94.42%.

Among the ensemble learning algorithms, the results remained highly competitive. Random Forest achieved an Accuracy of 91.63%, a Precision of 92.59%, a Recall of 91.63%, an F1-Score of 91.52%, and a ROC-AUC of 94.86%. XGBoost achieved an Accuracy of 91.63%, a Precision of 92.58%, a Recall of 91.63%, an F1-Score of 91.54%, and a ROC-AUC of 95.12%. LightGBM achieved the highest baseline performance, with an Accuracy of 91.70%, a Precision of 92.62%, a Recall of 91.70%, an F1-Score of 91.60%, and a ROC-AUC of 95.14%.

Overall, the baseline results show that the three ensemble learning models—XGBoost, LightGBM, and Random Forest—outperformed SVM and MLP across the evaluated metrics. Although LightGBM achieved the highest overall baseline performance, these results were used as a **reference point for the subsequent feature engineering stage**, rather than as the basis for final model selection. Following the baseline experiments, Mutual Information analysis and feature engineering were performed, resulting in a refined feature representation of **32 features**. Subsequently, **SVM, MLP, and XGBoost** were selected for further experimentation using the engineered feature set.

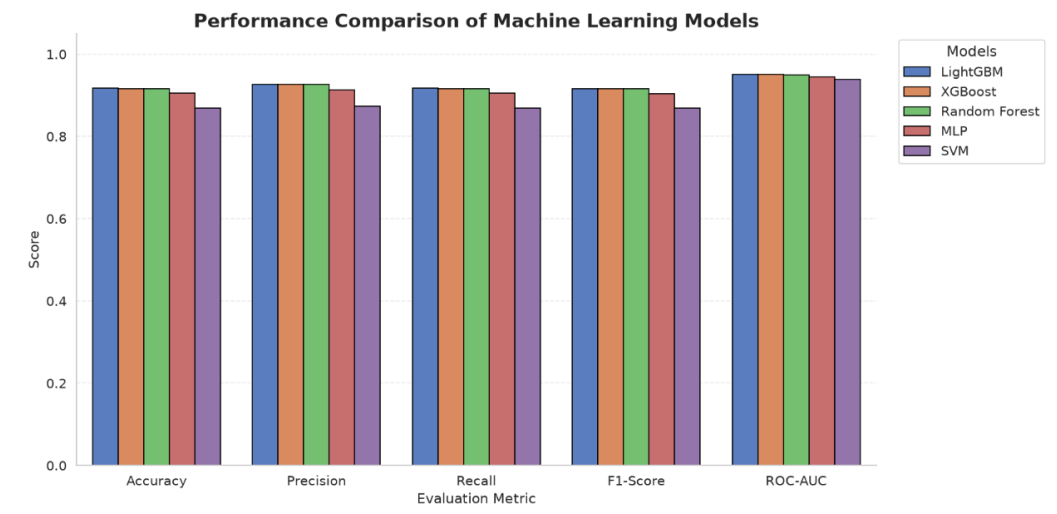


Fig. (4.8) Performance Comparison of Machine Learning Models

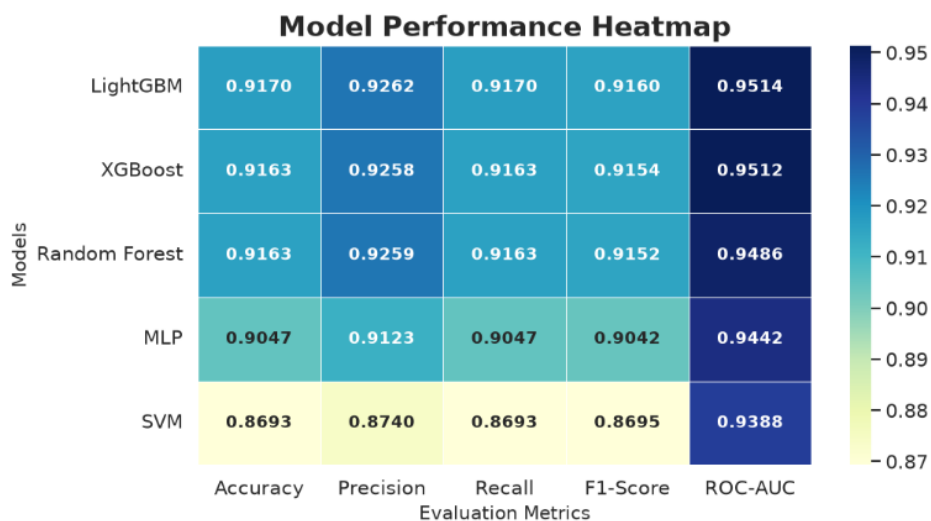


Fig. (4.9) Heatmap Comparison of Machine Learning Evaluation Metrics

Following the baseline comparison, three models—Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and XGBoost—were selected for the subsequent feature engineering experiments. The selection was based on model diversity rather than baseline performance alone. SVM represents margin-based learning, MLP represents neural-network-based nonlinear learning, while XGBoost represents tree-based ensemble learning. This diversity enabled the impact of the engineered feature representation to be evaluated across different learning paradigms.

4.6 Model Optimization

4.6.1 Feature Engineering

Following the baseline modeling experiments, additional feature engineering techniques were investigated to determine whether clinically meaningful derived variables could improve the predictive performance of the proposed diabetes prediction framework.

Several engineered features were evaluated, including the **BMI-to-Waist-Hip Ratio**, **Total Cholesterol-to-HDL Ratio**, **Triglyceride-Glucose (TyG) Index**, and **Sleep-to-Screen Time Ratio**. These features were designed to represent different clinical and lifestyle relationships, including central obesity, cardiometabolic risk, insulin resistance, and the balance between sleep and sedentary behavior.

Each engineered feature was incorporated into the modeling pipeline and evaluated under the same experimental conditions. However, the experimental results showed that these additional features **did not improve predictive performance**. In fact, their inclusion resulted in a decrease in the overall model performance compared with the corresponding models without these features.

Therefore, these engineered features were **excluded from the final feature representation**. This experimental step demonstrated that adding clinically meaningful derived variables does not necessarily improve predictive performance, as their information may already be sufficiently represented by the original features or may introduce additional complexity without providing useful predictive signal.

Following these experiments, **Mutual Information analysis** was used to further investigate the information content of the available features. Based on this analysis, nine low-information features were removed, while the **Postprandial-to-Fasting Glucose Ratio** (`postprandial_to_fasting_ratio`) was introduced as the final engineered feature.

The removal of the nine low-information features reduced the feature space from **40 to 31 features**, while the addition of the new glucose ratio resulted in a final feature representation of **32 features**. This refined

feature set was subsequently used for the second-stage experiments involving **SVM, MLP, and XGBoost**.

4.6.2 Models evaluations after optimization

Following the initial baseline modeling stage, a feature engineering and selection process was conducted to investigate whether modifying the feature representation could improve predictive performance. The analysis included evaluating clinically motivated derived features and removing features that demonstrated limited contribution to the prediction task.

Several engineered features were initially investigated. However, some of the evaluated features did not improve model performance and, in certain experiments, resulted in a reduction in predictive quality. Therefore, these features were not retained in the final feature set. Based on the experimental results and Mutual Information analysis, nine low-information features were removed, while the **Postprandial-to-Fasting Glucose Ratio** was retained as a clinically meaningful engineered feature. This process reduced the feature representation from 40 one-hot encoded features to a final set of **32 features**.

Following this refinement, the three selected models—**Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and XGBoost**—were re-evaluated using the optimized feature representation.

Multi-Layer Perceptron (MLP)

The MLP model achieved an overall accuracy of **90.22%**, with a precision of **90.71%**, recall of **90.22%**, F1-score of **90.16%**, and ROC-AUC of **94.55%**. These results demonstrate that the MLP was able to capture nonlinear relationships within the optimized feature representation

XGBoost

XGBoost achieved the highest overall performance among the three evaluated models, with an accuracy of **91.61%**, precision of **92.56%**, recall of **91.61%**, F1-score of **91.52%**, and ROC-AUC of **95.14%**. These results indicate that XGBoost benefited from the optimized feature representation and maintained strong discriminative capability.

Support Vector Machine (SVM)

The SVM model achieved an accuracy of **86.13%**, precision of **86.19%**, recall of **86.13%**, F1-score of **86.09%**, and ROC-AUC of **93.73%**. Although SVM maintained reasonable predictive performance, its results remained lower than those of XGBoost and MLP.

Conclusion

The experimental results showed that **XGBoost achieved the best overall performance among the three models after feature engineering and optimization**, followed by MLP and SVM. These results supported the continued use of XGBoost as a strong candidate for the subsequent stages of the proposed diabetes prediction framework. All experimental runs and corresponding evaluation metrics were recorded using **MLflow** to support systematic comparison and reproducibility.

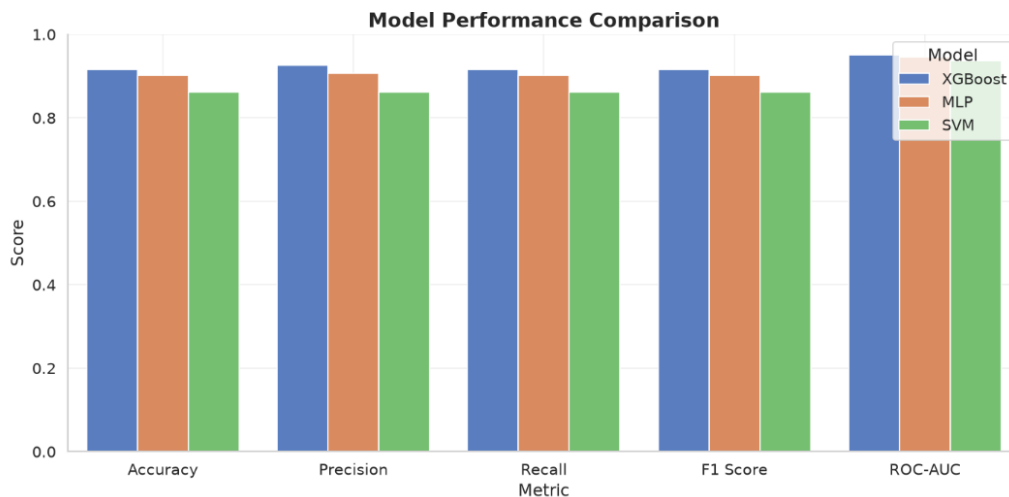


Fig. (4.10) Model Performance Comparison After Feature Engineering

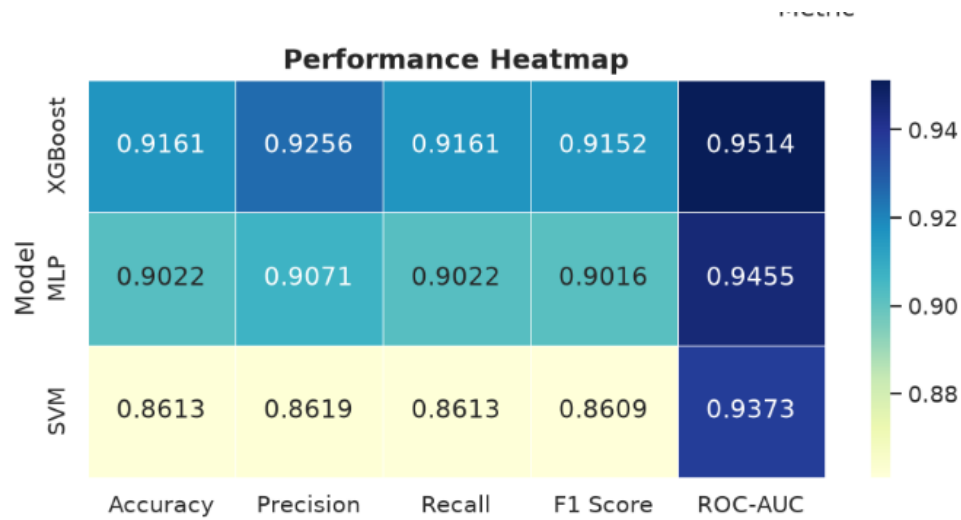


Fig. (4.11) Heatmap Representation of Model Performance Metrics After Feature Engineering

4.6.3 PCA Result

Principal Component Analysis (PCA) was investigated as a dimensionality reduction technique to examine whether the optimized feature space could be represented using a smaller number of components while preserving most of the information contained in the original features.

Following the feature engineering and selection stage, the dataset contained **32 features**. PCA was subsequently applied to the training data, and the resulting transformation was applied to the validation and test sets. Using a cumulative explained variance threshold of approximately **95%**, PCA reduced the original **32-dimensional feature space to only 3 principal components**, while retaining **95.57% of the total variance**.

The PCA-transformed dataset was then used to train and evaluate the three selected models:

Support Vector Machine (SVM), XGBoost, and Multi-Layer Perceptron (MLP).

ROC-AUC	F1-Score	Recall	Precision	Accuracy	Model
0.9156	0.8206	0.8205	0.8224	0.8205	MLP
0.9150	0.8201	0.8200	0.8217	0.8200	XGBoost
0.9132	0.8147	0.8150	0.8152	0.8150	SVM

Fig. (4.12) Model Performance After PCA

The results show that **MLP achieved the highest performance after PCA**, with an Accuracy of **82.05%** and a ROC-AUC of **91.56%**, followed closely by XGBoost and SVM.

However, when compared with the models trained using the optimized **32-feature representation without PCA**, the PCA-based representation resulted in lower predictive performance. This indicates that, although the three principal components retained **95.57% of the overall variance**, the dimensionality reduction also removed or compressed information that was useful for the classification task.

Therefore, PCA was **not adopted as the final feature representation for model training and deployment**. Instead, it was retained as an analytical technique to investigate the structure and dimensionality of the feature space.

4.6.4 Neural Network (NN)

A feed-forward Neural Network (NN) was implemented using TensorFlow and Keras to investigate the capability of a deeper neural architecture in modeling the complex nonlinear relationships among the clinical and lifestyle features. The network consisted of three hidden layers with 64, 32, and 16 neurons, respectively, using ReLU activation functions. Batch normalization, dropout, and L2 regularization were incorporated to improve training stability and reduce overfitting. The output layer used the Softmax activation function to perform multi-class diabetes classification.

The model was trained using the feature-engineered dataset with the Adam optimizer and a learning rate of 0.0007. Early stopping and adaptive learning-rate reduction were employed to prevent unnecessary training and improve convergence. All training experiments were tracked using MLflow to ensure reproducibility and facilitate systematic evaluation.

The Neural Network achieved a validation Accuracy of 91.51%, Precision of 92.39%, Recall of 91.51%, F1-score of 91.42%, and ROC-AUC of 95.11%. These results demonstrate that the neural architecture was capable of capturing complex nonlinear patterns within the clinical data and achieved competitive predictive performance.

ROC-AUC	F1-Score	Recall	Precision	Accuracy	Model
95.11%	91.42%	91.51%	92.39%	91.51%	Neural Network (NN)
94.55%	90.16%	90.22%	90.71%	90.22%	MLP
0.56%+	1.26%+	1.29%+	1.68%+	1.29%+	Improvement

Fig. (4.13) Comparison Between Neural Network and MLP

The performance of the proposed Neural Network was compared with the previously evaluated Multi-Layer Perceptron (MLP) classifier. Although both models belong to the family of feed-forward neural networks, the implemented Neural Network employed a specifically designed architecture incorporating multiple regularization and optimization mechanisms, including Batch Normalization, Dropout, L2 regularization, Early Stopping, and adaptive learning-rate reduction.

The results show that the proposed Neural Network consistently outperformed the MLP across all evaluation metrics. It achieved a 1.29 percentage-point improvement in accuracy and recall, while precision increased by 1.68 percentage points. The F1-score also improved by 1.26 percentage points, indicating a better balance between precision and recall. Furthermore, the ROC-AUC increased from 94.55% to 95.11%, demonstrating improved discrimination between the diabetes classes.

The improvement can be attributed to the more carefully designed architecture of the Neural Network, which combines three hidden layers, Batch Normalization, Dropout, L2 regularization, and adaptive optimization strategies. These mechanisms allowed the network to learn more complex representations while controlling overfitting during training.

However, despite its strong performance, the Neural Network was not selected as the final model. The final model selection was based on the overall experimental comparison and the requirements of the proposed explainable system. XGBoost remained the selected final model, providing strong predictive performance together with direct compatibility with SHAP-based feature-level explanations and practical deployment.

4.7 Best Performing Model

The final stage of the experimental evaluation involved selecting the most appropriate model for the proposed explainable diabetes prediction framework. Model selection was based primarily on predictive performance obtained during the experimental evaluation and optimization stages, while also considering the model's suitability for explainability and subsequent deployment.

Following the initial baseline evaluation of five machine learning models, **Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and XGBoost** were selected for further experimentation. Among these models, **XGBoost demonstrated the best overall predictive performance**, achieving an Accuracy of **91.61%**, Precision of **92.56%**, Recall of **91.61%**, F1-score of **91.52%**, and ROC-AUC of **95.14%**.

Comparison Between Neural Network and XGBoost

To further evaluate the proposed Neural Network, its performance was compared with the selected XGBoost model using both predictive and computational metrics. The results were obtained from the latest MLflow runs under the same feature-engineering setting.

```
=====
COMPARISON TABLE: Neural Network VS XGBoost (Latest Runs)
=====
runName      training_time_sec  inference_time_sec  trainable_parameters  accuracy  f1_score  roc_auc
NeuralNetwork_Multiclass_After_FE      510.022161          0.415632             5189    0.915133  0.914158  0.951069
XGBoost_Multiclass_After_FE            2.734591           0.091238            83908    0.916133  0.915194  0.951366
=====

SUMMARY OF DIFFERENCES (XGBoost vs Neural Network):
Training is faster in XGBoost by 507.29 sec
Inference is faster in XGBoost by 0.324395 sec
Neural Network has fewer parameters (lighter) by 78,719 params
XGBoost is more accurate by 0.10%

Comparison completed using logged MLflow data.
```

Fig. (4.14) Quantitative Comparison Between Neural Network and XGBoost Models

The results show that XGBoost achieved slightly higher predictive performance, with a 0.10 percentage-point improvement in Accuracy, while also achieving marginally higher F1-Score and ROC-AUC.

A more significant difference was observed in computational efficiency. XGBoost required only 2.48 seconds for training compared with 462.07 seconds for the Neural Network, making it approximately 186 times faster during training. XGBoost was also faster during inference, requiring only 0.037 seconds compared with 0.419 seconds for the Neural Network.

Although the Neural Network had substantially fewer trainable parameters (5,189 vs. 83,908), this did not translate into faster training or inference. Therefore, considering the slightly superior predictive performance, substantially faster execution, and compatibility with SHAP-based explainability, XGBoost was retained as the final model for the proposed diabetes prediction system.

Final Input Features Used by XGBoost

The final XGBoost model used in the proposed diabetes prediction system was trained using **12 clinical and demographic input features**. These features were:

- Age
- Gender
- Height (cm)
- Weight (kg)

- **HbA1c (%)**
- **Fasting Glucose (mg/dL)**
- **Postprandial Glucose (mg/dL)**
- **Total Cholesterol (mg/dL)**
- **Systolic Blood Pressure (mmHg)**
- **Family History of Diabetes**
- **Physical Activity (min/week)**
- **Insulin Level (μ IU/mL)**

These features represent the patient information required by the final prediction model and correspond to the clinical variables collected through the proposed web-based system.

The feature engineering and feature-selection experiments described in the preceding sections were used to investigate alternative feature representations and their effect on predictive performance. However, the final XGBoost model selected for deployment was trained using the **12 input features listed above**.

XGBoost Explainability Using SHAP

Following the selection of XGBoost as the final prediction model, **SHAP (SHapley Additive exPlanations)** was used to interpret the model predictions. The purpose of SHAP was to provide insight into how the input features contributed to the predictions generated by the trained XGBoost model.

SHAP analysis provides two complementary levels of interpretation. **Global interpretation** was used to examine the overall importance of features across the model, while **local interpretation** was used to examine the contribution of individual features to a specific patient prediction.

The SHAP analysis therefore represents a **model-specific interpretation stage rather than a feature-selection method**. It explains the behavior of the final trained XGBoost model based on the input features used by that model.

Final Model Integration

The selected XGBoost model was subsequently integrated into the proposed web-based diabetes risk assessment system. The system collects the required patient information through the web interface and passes the input data to the backend prediction pipeline, where the trained XGBoost model generates the diabetes prediction.

The prediction output is then presented through the application together with the corresponding explanation component, providing additional insight into the factors contributing to the model's prediction.

Therefore, **XGBoost** was selected as the final model based on its superior predictive performance among the models evaluated during the optimization stage, together with its suitability for SHAP-based interpretation and practical deployment.

4.8 Web-Based Diabetes Risk Assessment System:

4.8.1 Web Application Overview

A web-based application was developed to provide an interactive interface for the proposed diabetes risk assessment system. The website acts as the integration layer between the trained machine learning model and the end user, allowing patient data to be entered manually or extracted automatically from medical laboratory reports using Optical Character Recognition (OCR).

The web application follows a client-server architecture in which the frontend is responsible for user interaction and visualization, while the backend handles data processing, OCR operations, model inference, and communication with the trained prediction model.

The overall workflow can be summarized as:

User Input → Frontend → Backend API → Data Processing → ML Model → Prediction → Result Visualization

The system also provides a dedicated doctor interface through which previously assessed patients, risk categories, statistical summaries, and clinical information can be reviewed.

4.8.2 System Architecture

The implemented system consists of three main layers:

Frontend Layer

The frontend provides the graphical user interface through which users interact with the system. It was implemented using React with TypeScript.

The frontend is responsible for:

- Collecting patient demographic information.

- Collecting clinical measurements.
- Collecting medical history information.
- Uploading medical laboratory reports.
- Displaying OCR-extracted laboratory values.
- Allowing users to complete missing information manually.
- Sending patient data to the backend.
- Displaying prediction results and confidence scores.
- Providing a doctor dashboard for reviewing patient assessments.

The interface was designed as a multi-step form to organize the required clinical information into logical categories.

The patient form is divided into three main steps:

1. Patient Information
2. Clinical Measurements
3. Medical History

4.8.3 Patient Data Collection

The system collects the clinical variables required by the trained prediction model.

Demographic and Anthropometric Information

The first stage collects:

- Gender
- Age
- Height
- Weight

The system automatically calculates the patient's Body Mass Index (BMI) from height and weight using:

$$\text{BMI} = \text{Weight} / \text{Height}^2$$

where height is converted from centimeters to meters before performing the calculation.

The calculated BMI is also classified into standard categories such as underweight, normal weight, overweight, and obese.

4.8.4 Clinical Measurements

The second stage collects the main laboratory and physiological measurements used by the prediction model.

These include:

- HbA1c
- Fasting blood glucose
- Postprandial blood glucose
- Total cholesterol
- Systolic blood pressure
- Insulin level

Input validation was implemented to ensure that entered values fall within predefined physiological and technical ranges.

For example, HbA1c is restricted to a valid numerical range, while age is additionally validated as an integer.

This validation mechanism prevents invalid or incomplete data from being submitted to the prediction API.

4.8.5 Medical History

The third stage collects additional patient-related risk factors.

The implemented interface includes:

- Family history of diabetes
- Physical activity
- Hypertension
- Heart disease
- Previous diabetes diagnosis

Boolean values selected by the user are converted into the numerical representation expected by the machine learning model.

For example:

Selected → 1

Not selected → 0

Physical activity is also converted into the corresponding weekly activity value used by the prediction model.

4.8.6 Machine Learning Model Integration

The trained machine learning model is integrated into the web application through a backend API.

The frontend does not directly access the machine learning model. Instead, it constructs a structured prediction request and sends it to the backend.

The prediction payload contains the required model features, including:

- age
- gender
- bmi
- hba1c
- glucose_fasting
- glucose_postprandial
- cholesterol_total
- systolic_bp
- family_history_diabetes
- physical_activity_minutes_per_week
- insulin_level

The backend receives this structured data, performs the required preprocessing, and passes the processed features to the trained model.

The resulting prediction is then returned to the frontend as a structured response.

This separation improves modularity and prevents the machine learning implementation from being directly exposed to the client side.

4.8.7 Backend Implementation

The backend was implemented using FastAPI, providing RESTful API endpoints for communication between the frontend and the machine learning system.

The backend acts as an intermediary between the web interface and the trained prediction model.

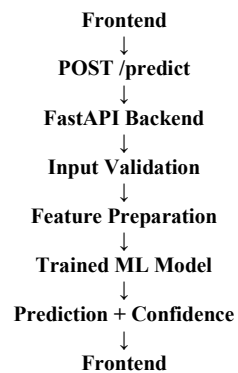
Its main responsibilities include:

- Receiving patient data.
- Validating API requests.

- Preparing model features.
- Executing model inference.
- Returning prediction results.
- Processing uploaded medical reports.
- Performing OCR.
- Parsing extracted laboratory values.
- Returning structured laboratory data to the frontend.

The main API operations include a prediction endpoint and an OCR endpoint.

The prediction workflow can be represented as:

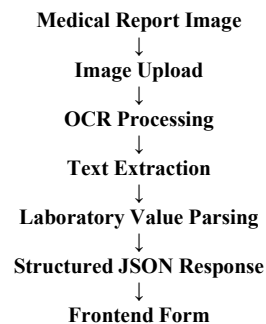


4.8.8 OCR-Based Medical Report Processing

To improve usability, an alternative input method was implemented for users who already have a medical laboratory report.

Instead of manually entering every laboratory value, the user can upload an image of the report.

The workflow is:



The OCR pipeline uses Tesseract OCR to recognize text contained in the uploaded medical laboratory report.

The extracted text is then processed by the OCR parser to identify relevant laboratory measurements.

The parser specifically searches for values associated with the required clinical variables, such as:

- HbA1c
- Fasting glucose
- Postprandial glucose
- Total cholesterol
- Insulin level

The extracted information is returned to the frontend as structured data rather than raw OCR text.

4.8.9 OCR and Manual Data Integration

An important design decision was to avoid automatically assuming values that were not detected by OCR.

Only laboratory values that are successfully detected are inserted into the existing form.

For example, if the uploaded report contains:

- HbA1c = 6.8
- Fasting Glucose = 125
- Cholesterol = 210

but does not contain insulin level, the system fills the detected values while leaving the insulin field empty.

The user is then asked to complete the missing information manually.

This hybrid approach combines the convenience of automated extraction with human verification and prevents unsupported assumptions from being introduced into the prediction pipeline.

4.8.10 OCR Error Handling

The OCR system also includes error-handling mechanisms.

If the image cannot be processed, the frontend displays an appropriate error message rather than allowing invalid OCR results to enter the prediction pipeline.

During development, the OCR module was configured to use the Tesseract executable installed on the local machine.

The OCR component therefore requires Tesseract OCR to be properly installed and available through the system environment or explicitly configured in the backend.

4.8.11 Prediction Result Visualization

After the backend completes model inference, the prediction response is displayed through a dedicated results interface.

The result interface presents:

- Predicted diabetes category
- Risk level
- Model confidence
- Clinical interpretation
- Patient-related information
- Visual indicators for easier interpretation

The model confidence is represented using a semi-circular gauge chart.

The confidence value is normalized to a range between 0 and 100 and displayed as a percentage.

This visualization allows the user to quickly understand the model's confidence in its prediction without requiring interpretation of raw numerical outputs.

4.8.12 Explanation and Interpretability

The prediction results also include an explanation component intended to make the model output easier to understand.

Instead of presenting only the predicted class, the system provides an interpretation of the patient's clinical measurements and their relationship to the resulting risk assessment.

This improves the transparency of the system and makes the prediction more understandable to users.

The explanation layer is presented separately from the core prediction result so that the actual machine learning prediction remains distinguishable from its interpretation.

4.8.13 Doctor Dashboard

A dedicated doctor dashboard was implemented to provide a clinical overview of previously assessed patients.

The dashboard retrieves patient records from the backend and summarizes them into several categories.

The main dashboard statistics include:

- Total number of patients
- Healthy patients
- Pre-diabetes cases
- Type 2 diabetes cases
- High-risk patients

The dashboard also provides visual charts for analyzing the collected patient population.

These include:

- Prediction distribution
- Age distribution
- HbA1c distribution
- Monthly assessment/report statistics

This provides a high-level overview of the patient population and allows healthcare professionals to identify important trends.

4.8.14 High-Risk Patient Monitoring

A dedicated section of the doctor dashboard displays patients classified as high risk.

The system filters patient records according to the prediction risk level and displays the most relevant high-risk cases.

Each patient entry provides basic information such as:

- Patient name
- HbA1c level
- Predicted diabetes category
- Risk classification

Selecting a patient opens a detailed clinical information drawer.

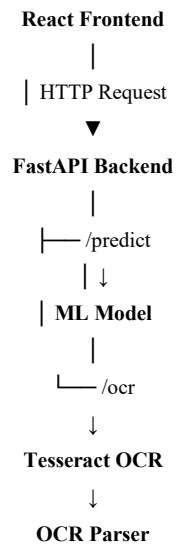
This design allows healthcare professionals to quickly identify patients who may require further clinical review.

4.8.15 Frontend–Backend Communication

Communication between the React frontend and FastAPI backend is performed through HTTP requests.

For prediction, the frontend constructs the validated patient payload and sends it to the backend API.

For OCR processing, the uploaded image is sent using FormData as a multipart HTTP request. Conceptually:



The backend returns structured JSON responses that are consumed by the frontend components.

This architecture allows the frontend and backend to remain independent and makes it easier to replace or update the machine learning model without redesigning the user interface.

4.8.16 Input Validation and Error Handling

Several validation mechanisms were implemented throughout the application.

The frontend uses a schema-based validation approach to ensure:

- Required fields are completed.
- Numeric fields contain valid numbers.
- Values fall within acceptable ranges.
- Age is an integer.
- Gender is selected.
- Boolean medical-history fields have defined values.

Errors are displayed directly beside the relevant input fields.

The application also handles backend and network failures by displaying an analysis failure state with the corresponding error information.

This prevents the user from receiving an ambiguous result when the prediction service is unavailable.

4.8.17 RAG Integration with the Web-Based Platform

To extend the platform beyond numerical risk prediction, the adapted RAG pipeline described in Section 3.7 was integrated into the web application as a dedicated nutrition question-answering feature. Rather than remaining an isolated offline component, the pipeline was exposed to end users through a chatbot-style interface embedded within the platform, allowing patients and clinicians to ask free-text questions related to diabetes nutrition management directly within the assessment environment.

When a query is submitted, the same **hybrid retrieval, cross-encoder reranking, and grounded-generation** stages described previously are applied to the *Nutritional Management of Diabetes Mellitus* reference. The generated response is then presented to the user together with the corresponding confidence label and cited source pages, allowing the supporting information to be traced back to the reference document.

The RAG component complements the predictive and explainability components of the system. While the **XGBoost model** generates the diabetes prediction and **SHAP** explains the contribution of the input features to that prediction, the RAG module provides additional evidence-based nutritional information in response to user questions. Importantly, the RAG component does not modify or determine the output of the XGBoost prediction model.

This integration transforms the website from a standalone prediction interface into an integrated decision-support platform that combines **machine learning-based risk assessment, model explainability, and retrieval-grounded nutritional information**. The use of retrieved source content also improves the traceability of the generated responses and reduces reliance on unsupported generative outputs.

4.8.18 Personalized Recommendations

To complement the diabetes risk prediction and SHAP-based explanation, the web application provides personalized preventive recommendations based on the user's health profile and identified risk factors. The recommendations are designed to translate the prediction results into practical lifestyle guidance rather than presenting the risk assessment as a numerical result only.

The recommendation component considers relevant clinical and lifestyle characteristics available in the patient's assessment, including factors such as BMI, physical activity, smoking status, sleep habits, and other relevant health indicators. Based on the identified risk factors, the system presents appropriate preventive guidance related to healthy lifestyle practices, physical activity, smoking reduction, sleep improvement, and dietary awareness.

The recommendations are intended for awareness and preventive healthcare support and do not constitute medical diagnosis or treatment. They are presented alongside the prediction and explanation results to help users better understand potential actions associated with their identified risk profile.

4.8.19 Lifestyle Simulation and Counterfactual Risk Analysis

The web application also provides a lifestyle simulation feature that allows users to explore how changes in selected health and lifestyle factors may affect the predicted diabetes risk. Instead of modifying the original patient assessment, the simulation evaluates alternative input scenarios and compares the resulting prediction with the original risk assessment.

Users can modify selected behavioral or health-related parameters, such as BMI, smoking behavior, physical activity, and sleep-related factors, depending on the available input variables. The modified values are submitted through the same prediction pipeline, allowing the trained model to estimate the corresponding risk under the simulated scenario.

This functionality provides a simple form of counterfactual risk analysis by showing users how changes in modifiable factors may be associated with a different model prediction. The purpose of the simulation is educational and preventive, helping users understand the potential relationship between lifestyle changes and predicted diabetes risk. It does not represent a clinical prediction of future disease outcomes.

4.8.20 User Interface Design

The interface was designed using a modern medical dashboard style with emphasis on:

- Clear visual hierarchy
- Responsive layout
- Step-based navigation
- Consistent medical color coding
- Interactive cards
- Animated transitions
- Visual risk indicators
- Accessible form controls

The application uses reusable React components for elements such as:

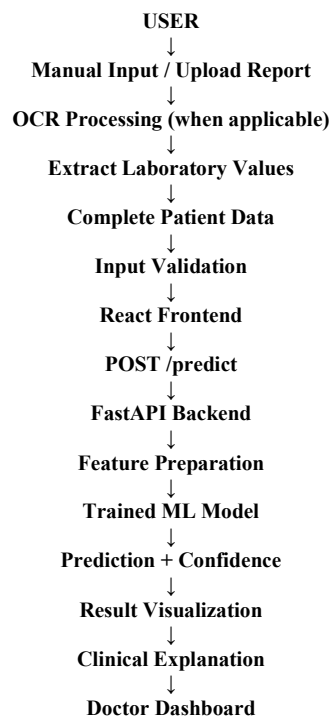
- Input fields

- Clinical measurement fields
- Statistic cards
- Charts
- Patient tables
- Patient detail drawers
- Prediction result components
- Confidence gauges

Component-based architecture reduces code duplication and improves maintainability.

4.8.21 Overall System Workflow

The complete implemented workflow can be summarized as follows:



4.8.22 Implementation Summary

The developed web application provides an end-to-end interface for deploying the proposed diabetes risk assessment model in a practical environment. The system integrates manual clinical data entry,

automated laboratory report processing through OCR, backend model inference, prediction visualization, and doctor-oriented patient monitoring within a single platform.

The separation between the frontend, backend, OCR pipeline, and machine learning model provides a modular architecture that facilitates maintenance and future development. In addition, the combination of automated extraction and manual verification reduces the dependence on fully automated OCR interpretation while preserving the convenience of image-based data entry.

Overall, the implemented website transforms the trained machine learning model from an isolated experimental component into an interactive decision-support application capable of receiving clinical data, generating predictions, and presenting the results in a structured and interpretable manner.

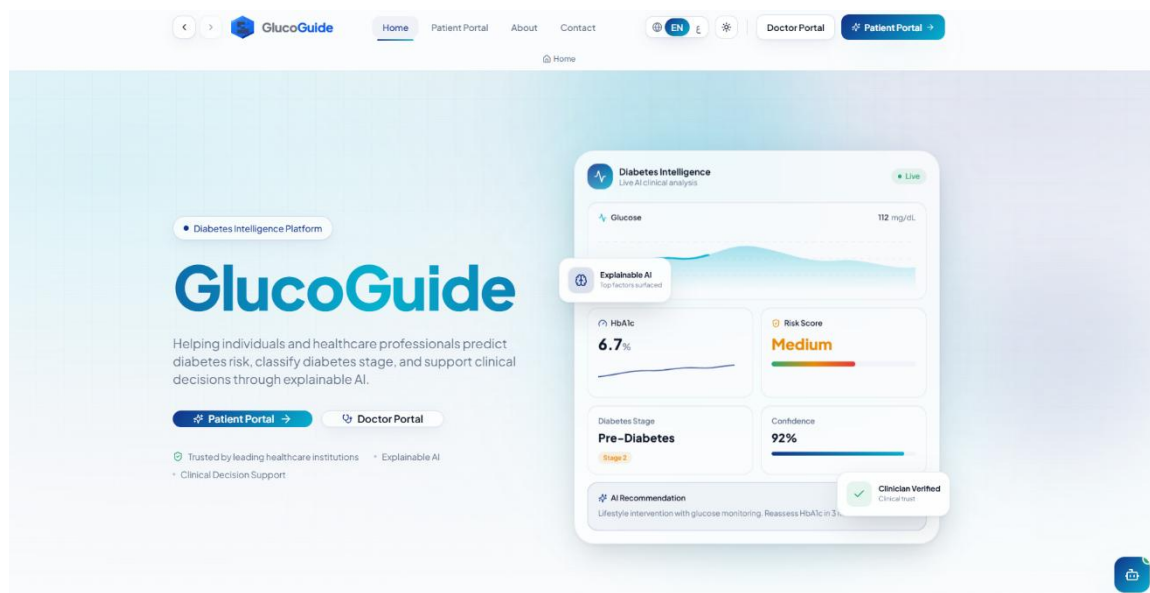


Fig. (4.15) User Interface Overview of the GlucoGuide Web Platform

CHAPTER 5

Conclusion and Future Work

5.1 Conclusion

The project aimed to develop an explainable and practical web-based diabetes risk assessment system that combines machine learning, explainable artificial intelligence, retrieval-augmented generation, and web technologies to support diabetes risk assessment and provide evidence-based guidance.

A structured machine learning pipeline was developed, including data preprocessing, feature engineering, feature selection, class balancing, and comparative model evaluation. Several machine learning approaches were investigated, including Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), Random Forest, XGBoost, and LightGBM. Feature engineering was explored using clinically motivated derived variables, while Mutual Information was employed to identify and remove low-information features. PCA was also evaluated as a dimensionality-reduction technique; however, its lower predictive performance demonstrated that dimensionality reduction was not beneficial for the final model. Therefore, the final prediction model was trained using the selected feature representation without PCA.

Among the evaluated models, XGBoost was selected as the final prediction model based on its strong predictive performance and practical efficiency. In the optimized evaluation, XGBoost achieved an Accuracy of 91.61%, Precision of 92.56%, Recall of 91.61%, F1-score of 91.52%, and ROC-AUC of 95.14%. In addition, the comparison with the implemented neural network showed that XGBoost achieved slightly higher predictive performance while requiring substantially less training and inference time. These results supported its selection as the most appropriate model for the final system.

To improve transparency, SHAP (SHapley Additive exPlanations) was integrated with the selected XGBoost model. SHAP provides both global and local explanations, allowing the system to identify the features that most influence the model's predictions and to explain the contribution of individual patient features to a specific risk assessment. This makes the prediction process more interpretable than presenting the model output alone.

The project was further extended through the integration of an Adapted Retrieval-Augmented Generation (RAG) pipeline for diabetes nutrition-related questions. The RAG component retrieves relevant information from a dedicated clinical reference, applies hybrid retrieval and reranking, and generates grounded responses based on the retrieved evidence. This allows the system to complement the numerical risk assessment with evidence-based nutritional guidance while maintaining a clear distinction between model prediction and generated informational responses.

Finally, the trained model and supporting components were integrated into a web-based diabetes risk assessment platform using a client-server architecture. The system provides manual patient data entry, medical laboratory report processing through OCR, automated extraction of laboratory values, input validation, prediction generation, SHAP-based interpretation, nutrition-oriented question answering, and a dedicated doctor dashboard for reviewing patient assessments and risk categories. The integration of these components transforms the machine learning models from isolated experimental implementations into an interactive decision-support prototype.

Overall, the project demonstrates that combining machine learning, explainable AI, retrieval-grounded knowledge assistance, and practical web deployment can provide a more comprehensive diabetes risk assessment platform. The proposed system is intended as a decision-support and risk-assessment tool rather than a replacement for professional medical diagnosis or clinical judgment.

5.2 Future Work

Although the proposed system achieved its main objectives, several extensions can be developed to transform the current prototype into a more comprehensive diabetes monitoring and clinical decision-support platform.

1. Integration with Continuous Glucose Monitoring Devices

A major future direction is to connect the system with Continuous Glucose Monitoring (CGM) devices and other glucose-monitoring technologies. Instead of relying only on manually entered laboratory measurements, the system could receive periodic or continuous glucose readings and analyze changes in glucose patterns over time. This would allow the prediction component to move from a single-point risk assessment toward longitudinal diabetes risk monitoring.

2. Integration with Clinics and Healthcare Professionals

The current doctor dashboard could be extended into a complete clinical monitoring interface for physicians and healthcare centers. Doctors could monitor registered patients, review their previous

assessments, compare changes in risk over time, and examine the clinical features contributing to each prediction through SHAP explanations. The system could also provide alerts when a patient's risk profile shows a significant deterioration, allowing the physician to determine whether further clinical assessment is required.

3. Longitudinal Risk Tracking

The current prediction represents an assessment based on the available patient information at a particular point in time. Future development could introduce time-series risk tracking, where multiple assessments are stored and compared to identify whether the patient's predicted risk is improving, remaining stable, or increasing. Visualization of risk trajectories could provide physicians with a clearer picture of patient progression.

4. Expansion of the RAG Knowledge Base

The current RAG component is based on a dedicated diabetes nutrition reference. Future versions could incorporate multiple authoritative clinical guidelines and nutritional references covering diabetes management, physical activity, nutrition, glucose monitoring, and preventive care. The retrieval system could also provide stronger source attribution by displaying the reference, section, and page associated with each generated response.

5. Improved OCR and Laboratory Report Processing

The existing OCR pipeline could be expanded to support a wider range of laboratory report layouts, languages, and image qualities. Future versions could incorporate stronger document-layout analysis and validation mechanisms to improve the extraction of laboratory values and reduce errors caused by unclear images, different report formats, or ambiguous laboratory units.

6. Integration with Electronic Health Records

The platform could eventually be connected to Electronic Health Record (EHR) systems, where appropriate and subject to healthcare interoperability and privacy requirements. This would reduce repeated manual data entry and allow authorized healthcare professionals to access relevant historical clinical information when performing a risk assessment.

7. Model Recalibration and Continuous Learning

As additional real-world patient data become available, future versions could investigate controlled model updating and recalibration. Instead of retraining the model automatically, new data could first undergo quality assessment and clinical validation before being incorporated into future model versions. This would help maintain model performance as patient populations and data distributions change.

References:

- [1] U. M. Butt, S. Letchmunan, M. Ali, F. H. Hassan, A. Baqir, and H. H. R. Sherazi, "Machine Learning Based Diabetes Classification and Prediction for Healthcare Applications," *J. Healthc. Eng.*, vol. 2021, pp. 1–17, Sep. 2021, doi: 10.1155/2021/9930985.
- [2] O. Iparraguirre-Villanueva, K. Espinola-Linares, R. O. Flores Castañeda, and M. Cabanillas-Carbonell, "Application of Machine Learning Models for Early Detection and Accurate Classification of Type 2 Diabetes," *Diagnostics*, vol. 13, no. 14, p. 2383, Jul. 2023, doi: 10.3390/diagnostics13142383.
- [3] T. Zhu, K. Li, P. Herrero, and P. Georgiou, "Deep Learning for Diabetes: A Systematic Review," *IEEE J. Biomed. Health Inform.*, vol. 25, no. 7, pp. 2744–2757, Jul. 2021, doi: 10.1109/JBHI.2020.3040225.
- [4] T. Sharma and M. Shah, "A comprehensive review of machine learning techniques on diabetes detection," *Vis. Comput. Ind. Biomed. Art*, vol. 4, no. 1, p. 30, Dec. 2021, doi: 10.1186/s42492-021-00097-7.
- [5] H. B. Kibria, M. Nahiduzzaman, Md. O. F. Goni, M. Ahsan, and J. Haider, "An Ensemble Approach for the Prediction of Diabetes Mellitus Using a Soft Voting Classifier with an Explainable AI," *Sensors*, vol. 22, no. 19, p. 7268, Sep. 2022, doi: 10.3390/s22197268.
- [6] U. Allani, "Interactive Diabetes Risk Prediction Using Explainable Machine Learning: A Dash-Based Approach with SHAP, LIME, and Comorbidity Insights," 2025, *arXiv*. doi: 10.48550/ARXIV.2505.05683.
- [7] A. Weaver, "Machine Learning and Evaluation of Diabetes Risk Using 2022 BRFSS Data," Master's Thesis, Western Michigan Univ., 2024. [Online]. Available: https://scholarworks.wmich.edu/masters_theses/5447/
- [8] B. Nguyen and Y. Zhang, "A Comparative Study of Diabetes Prediction Based on Lifestyle Factors Using Machine Learning," Mar. 06, 2025, *arXiv*: arXiv:2503.04137. doi: 10.48550/arXiv.2503.04137.
- [9] Z. Rafie, M. S. Talab, B. E. Z. Koor, A. Garavand, C. Salehnasab, and M. Ghaderzadeh, "Leveraging XGBoost and explainable AI for accurate prediction of type 2 diabetes," *BMC Public Health*, vol. 25, no. 1, p. 3688, Oct. 2025, doi: 10.1186/s12889-025-24953-w.
- [10] M. Panda and S. R. Mahanta, "Explainable artificial intelligence for Healthcare applications using Random Forest Classifier with LIME and SHAP," 2023, *arXiv*. doi: 10.48550/ARXIV.2311.05665.
- [11] P. B. Khokhar, V. Pentangelo, F. Palomba, and C. Gravino, "Towards Transparent and Accurate Diabetes Prediction Using Machine Learning and Explainable Artificial Intelligence," 2025, *arXiv*. doi: 10.48550/ARXIV.2501.18071.
- [12] Md. M. Islam, H. R. Rifat, Md. S. B. Shahid, A. Akhter, M. A. Uddin, and K. M. M. Uddin, "Explainable Machine Learning for Efficient Diabetes Prediction Using Hyperparameter Tuning, SHAP Analysis, Partial Dependency, and LIME," *Eng. Rep.*, vol. 7, no. 1, p. e13080, Jan. 2025, doi: 10.1002/eng2.13080.
- [13] M. M. El Sherbiny, M. G. Abdel Fattah, A. H. Rabie, A. E. Taki Eldin, and H. E. Moustafa, "A Diabetes Mellitus Prediction Model Based on Supervised Machine Learning Techniques," *Int. J. Telecommun.*, vol. 05, no. 01, pp. 1–11, Mar. 2025, doi: 10.21608/ijt.2025.359269.1083.
- [14] G. R. Ashisha, X. A. Mary, E. G. M. Kanaga, J. Andrew, and R. J. Eunice, "Random Oversampling-Based Diabetes Classification via Machine Learning Algorithms," *Int. J. Comput. Intell. Syst.*, vol. 17, no. 1, p. 270, Nov. 2024, doi: 10.1007/s44196-024-00678-3.

- [15] S. Elluri and K. V. S. Abinay, “Enhanced Diabetes Prediction Using a Hybrid Machine Learning Framework with Feature Selection and Weighted Ensemble Classification,” 2025.
- [16] Z. Ullah *et al.*, “Detecting High-Risk Factors and Early Diagnosis of Diabetes Using Machine Learning Methods,” *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–10, Sep. 2022, doi: 10.1155/2022/2557795.
- [17] A. Haque, S. Islam, N. R. Mim, S. M. Meem, A. Saha, and R. Khan, “Automatic diabetes prediction with explainable machine learning techniques,” *IAES Int. J. Artif. Intell. IJ-AI*, vol. 14, no. 1, p. 408, Feb. 2025, doi: 10.11591/ijai.v14.i1.pp408-415.
- [18] M. S. Alzboon, M. Al-Batah, M. Alqaraleh, A. Abuashour, and A. F. Bader, “A Comparative Study of Machine Learning Techniques for Early Prediction of Diabetes,” in *2023 IEEE Tenth International Conference on Communications and Networking (ComNet)*, Nov. 2023, pp. 1–12. doi: 10.1109/ComNet60156.2023.10366688.
- [19] W. Hoyos, K. Hoyos, R. Ruiz, and J. Aguilar, “An explainable analysis of diabetes mellitus using statistical and artificial intelligence techniques,” *BMC Med. Inform. Decis. Mak.*, vol. 24, no. 1, p. 383, Dec. 2024, doi: 10.1186/s12911-024-02810-x.

ملخص المشروع

يُعدّ داء السكري أحد أكبر التحديات الصحية عالمياً، حيث يُسهم الكشف المبكر عن الفئات الأكثر عرضة للإصابة في توفير التدخل الطبي المناسب وتحسين النتائج الصحية. يقدم هذا المشروع نظام **Glucoguide**، وهو منصة طبية تفاعلية عبر الويب مدعومة بتقنيات الذكاء الاصطناعي، تُعنى بالتقييم الأولي لمخاطر الإصابة بالسكري ودعم اتخاذ القرارات السريرية.

يعتمد نظام **Glucoguide** على دمج نموذج تنبؤي قائم على التعلم الآلي مع واجهة إلكترونية تُسجل البيانات الديموغرافية والسريرية والنمط المعيشي والتاريخ الطبي للمريض؛ بما في ذلك مؤشر كتلة الجسم (BMI)، ونسبة الجلوكوز، والسكر التراكمي (HbA1c)، ومستويات الكوليسترول، وضغط الدم، والأنسولين. كما يتضمن النظام تقنية التعرف الضوئي على الحروف (OCR) لاستخراج النتائج التحليلية تلقائياً من التقارير الطبية المرفوعة، مما يُقلل من الإدخال اليدوي ويُرفع كفاءة التقييم.

تُعالج البيانات المُجمعة وتُمرر عبر واجهة برمجة التطبيقات التنبؤية (Prediction API) إلى نموذج التعلم الآلي المُدرّب، لإنتاج تقدير دقيق لنسبة مخاطر الإصابة. وتُعرض هذه النتائج عبر واجهة مستخدم سلسلة تُساعد في تفسير المخاطر وزيادة الوعي الصحي. ختاماً، يُبرهن نظام **Glucoguide** على فعالية دمج الذكاء الاصطناعي والتعلم الآلي وتقنيات الويب في تقديم تقييم صحي أولي يتسم بالسرعة وسهولة الوصول، مع التأكيد على أن النظام يُعد أداة مُساندة للتشخيص الطبي والتقدير السريري وليس بديلاً عنهما.

الكلمات المفتاحية: تقييم مخاطر السكري، الذكاء الاصطناعي، التعلم الآلي، التعرف الضوئي على الحروف (OCR)، دعم القرارات السريرية، الأنظمة الصحية الإلكترونية.



الجهة المانحة



صفحة الموافقة

التنبؤ بمرض السكري وتصنيفه باستخدام الذكاء الاصطناعي

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رسالة مقدمة استكمالاً لمتطلبات الحصول على

Digilians 9 Months دبلومة

مسار الذكاء الاصطناعي التطبيقي و تحليل البيانات

يعتمد من لجنة الممتحنين

التوقيع	الإسم
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الجهة المانحة



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رسالة مقدمة استكمالاً لمتطلبات الحصول على دبلومة Digilians لمدة تسعة أشهر في الذكاء الاصطناعي التطبيقي وتحليل البيانات

تحت إشراف

لواء أ.د / كمال الحداد

الكلية الفنية العسكرية

أغسطس 2026